Standard Model Signal in Cosmological Higgs Collider

Abstract

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1 Introduction

2 Gauge Boson Signal

2.1 Preliminaries

We will compute the connected four-point function $\langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle'$ generated by a massive gauge-boson loop. Our interest is the leading nonlocal signal in the collapsed s channel,

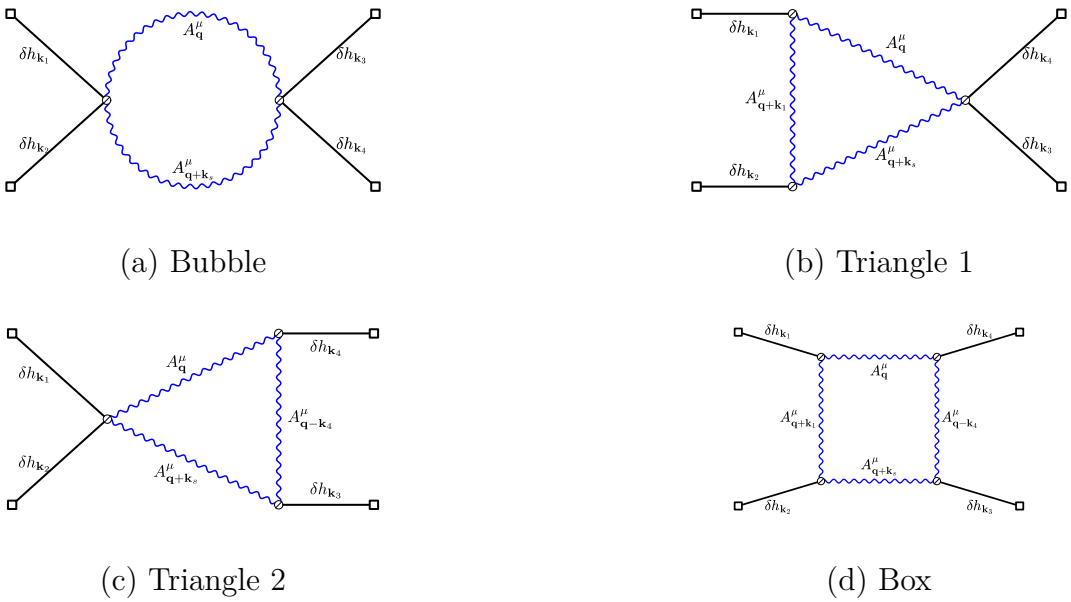
$$\mathbf{k}_s \equiv \mathbf{k}_1 + \mathbf{k}_2 = -(\mathbf{k}_3 + \mathbf{k}_4), \qquad k_s \ll k_i, \qquad k_{12} \equiv k_1 + k_2, \quad k_{34} \equiv k_3 + k_4, \tag{1}$$

where $k_i \equiv |\mathbf{k}_i|$. Thus $k_1 \simeq k_2$ and $k_3 \simeq k_4$ at leading order. We choose $\mathbf{k}_s$ along the z axis and set the azimuth of $\mathbf{k}_1$ to zero. Here θ_1 and θ_3 are the polar angles of $\mathbf{k}_1$ and $\mathbf{k}_3$, respectively, and ϕ_3 is their relative azimuth.

The Higgs–gauge couplings follow from the Higgs kinetic term [1]:

$$\frac{\sqrt{-g}}{2} |\mathrm{D}_\mu \mathbf{H}|^2 \supset \frac{a^2 g^2}{4} (h_0 + \delta h)^2 \left(W^+_\mu W^{-\mu} + \frac{Z_\mu Z^\mu}{2 c_W^2} \right), \tag{2}$$

where g is the weak gauge coupling, $c_W = \cos\theta_W$, and the Lorentz contractions on the right use the flat conformal metric. The terms linear and quadratic in δh generate the cubic and quartic vertices, respectively. They give the bubble, two triangle placements, and box shown in Fig. 1.



(a) Bubble (b) Triangle 1

(c) Triangle 2 (d) Box

Figure 1: The four gauge-boson loop diagrams considered here. Black solid lines denote external Higgs modes and blue wavy lines denote gauge-boson propagators.

In the light-Higgs regime, the dominant infrared Higgs-loop contribution to the vector masses is [1]

$$m_W^2 \simeq \frac{3 g^2 H^4}{8\pi^2 m_H^2}, \qquad m_Z^2 \simeq \frac{m_W^2}{c_W^2}. \tag{3}$$

Here m_H is the effective Higgs mass. We use a constant vector mass m_V and the parameter $\tilde{\nu} \equiv \sqrt{m_V^2/H^2 - 1/4} > 0$ in the following. The W and Z contributions are obtained with their respective masses and couplings in (2).

We use $\lambda, \sigma = L, +, -, T$ for vector polarization-basis labels, and $\eta = \pm 1$ for the two nonlocal branches. Neglecting the Higgs mass in the external mode functions, we write its bulk-to-boundary propagator as

$$G_{\mathsf{a}}(k;\tau) = \frac{H^2}{2k^3}(1 - \mathrm{i}\mathsf{a}k\tau)e^{\mathrm{i}\mathsf{a}k\tau}. \tag{4}$$

The spacetime vector propagator decomposes as

$$\begin{aligned} D_{\mathsf{ab},\mu\nu}(\mathbf{k};\tau_1,\tau_2) = &\sum_{\lambda=T,L,\pm} e^{(\lambda)}_{\mathbf{k},\mu} e^{(\lambda)*}_{\mathbf{k},\nu} D^{(\lambda)}_{\mathsf{ab}}(k;\tau_1,\tau_2) \\ &+ e^{(T)}_{\mathbf{k},\mu} e^{(L)*}_{\mathbf{k},\nu} D^{(TL)}_{\mathsf{ab}}(k;\tau_1,\tau_2) + e^{(L)}_{\mathbf{k},\mu} e^{(T)*}_{\mathbf{k},\nu} D^{(LT)}_{\mathsf{ab}}(k;\tau_1,\tau_2), \end{aligned} \tag{5}$$

where $\mu, \nu = 0, 1, 2, 3$. In the comoving frame, the covariant polarization basis is [2]

$$e^{(T)}_{\mathbf{k},\mu} = (1, \mathbf{0}), \qquad e^{(L)}_{\mathbf{k},\mu} = (0, \widehat{\mathbf{k}}), \qquad e^{(\pm)}_{\mathbf{k},\mu} = \frac{1}{\sqrt{2}}(0, \mathbf{e}_\theta \pm \mathrm{i}\mathbf{e}_\phi),$$

with $(\widehat{\mathbf{k}}, \mathbf{e}_\theta, \mathbf{e}_\phi)$ the orthonormal spherical basis. The label T denotes the temporal component of the longitudinal mode.

At zero chemical potential, all components can be expressed through an auxiliary scalar propagator with the same Hankel index [2, 3]:

$$D^{(\mathrm{s})}_{>}(k;\tau_1,\tau_2) = \frac{\pi H^2}{4} e^{-\pi\tilde{\nu}} (\tau_1\tau_2)^{3/2} \mathrm{H}^{(1)}_{\mathrm{i}\tilde{\nu}}(-k\tau_1) \mathrm{H}^{(2)}_{-\mathrm{i}\tilde{\nu}}(-k\tau_2), \tag{6}$$

$$D^{(\pm)}_{>} = \frac{D^{(\mathrm{s})}_{>}}{H^2\tau_1\tau_2}, \qquad D^{(L)}_{>} = \frac{(\partial_{\tau_1} - 2/\tau_1)(\partial_{\tau_2} - 2/\tau_2) D^{(\mathrm{s})}_{>}}{H^2(\tilde{\nu}^2 + 1/4)}, \tag{7}$$

$$D^{(T)}_{>} = \frac{k^2 D^{(\mathrm{s})}_{>}}{H^2(\tilde{\nu}^2 + 1/4)}, \tag{8}$$

$$D^{(TL)}_{>} = \frac{\mathrm{i}k(\partial_{\tau_2} - 2/\tau_2) D^{(\mathrm{s})}_{>}}{H^2(\tilde{\nu}^2 + 1/4)}, \qquad D^{(LT)}_{>} = -\frac{\mathrm{i}k(\partial_{\tau_1} - 2/\tau_1) D^{(\mathrm{s})}_{>}}{H^2(\tilde{\nu}^2 + 1/4)}. \tag{9}$$

With spacetime indices and momentum arguments suppressed, $D_{-+} = D_>$, $D_{+-} = D_<$, and the separated-time SK propagators are

$$\begin{aligned} D_{++} &= \theta(\tau_1 - \tau_2) D_> + \theta(\tau_2 - \tau_1) D_<, \\ D_{--} &= \theta(\tau_2 - \tau_1) D_> + \theta(\tau_1 - \tau_2) D_<. \end{aligned} \tag{10}$$

These relations apply to $D^{(\pm)}, D^{(L)}, D^{(TL)}$ and $D^{(LT)}$. But for $D^{(T)}$, the $++$ and $--$ propagators have a delta-function term because the temporal mode is not a physical degree of freedom:

$$\begin{aligned} D^{(T)}_{++} &= \theta(\tau_1 - \tau_2) D^{(T)}_{>} + \theta(\tau_2 - \tau_1) D^{(T)}_{<} + \frac{\mathrm{i}\delta(\tau_1 - \tau_2)}{H^2(\tilde{\nu}^2 + 1/4) a(\tau_1)^2}, \\ D^{(T)}_{--} &= \theta(\tau_2 - \tau_1) D^{(T)}_{>} + \theta(\tau_1 - \tau_2) D^{(T)}_{<} - \frac{\mathrm{i}\delta(\tau_1 - \tau_2)}{H^2(\tilde{\nu}^2 + 1/4) a(\tau_1)^2}. \end{aligned} \tag{11}$$

We will see that the propagators $D^{(T)}$, $D^{(TL)}$ and $D^{(LT)}$ will not contribute to the leading nonlocal signal.

The leading nonlocal parts of the spatial propagators are independent of the SK indices:

$$\left[D^{(\pm)}_{\mathsf{ab}}(k;\tau_1,\tau_2)\right]_{\rm NonL} = \frac{(\tau_1\tau_2)^{1/2}}{4\pi}\sum_{\eta=\pm1}\Gamma^2(-\mathrm{i}\eta\tilde{\nu})\left(\frac{k^2\tau_1\tau_2}{4}\right)^{\mathrm{i}\eta\tilde{\nu}}, \tag{12}$$

$$\left[D^{(L)}_{\mathsf{ab}}(k;\tau_1,\tau_2)\right]_{\rm NonL} = \frac{(\tau_1\tau_2)^{1/2}}{4\pi}\sum_{\eta=\pm1}\frac{(-1/2+\mathrm{i}\eta\tilde{\nu})^2}{\tilde{\nu}^2+1/4}\Gamma^2(-\mathrm{i}\eta\tilde{\nu})\left(\frac{k^2\tau_1\tau_2}{4}\right)^{\mathrm{i}\eta\tilde{\nu}}. \tag{13}$$

The temporal nonlocal part is

$$\left[D^{(T)}_{\mathsf{ab}}(k;\tau_1,\tau_2)\right]_{\rm NonL} = \frac{k^2(\tau_1\tau_2)^{3/2}}{4\pi(\tilde{\nu}^2+1/4)}\sum_{\eta=\pm1}\Gamma^2(-\mathrm{i}\eta\tilde{\nu})\left(\frac{k^2\tau_1\tau_2}{4}\right)^{\mathrm{i}\eta\tilde{\nu}}. \tag{14}$$

The one-loop factorization theorem and cutting rule [4] isolate the leading nonlocal signal in the collapsed channel. We first identify the two internal lines and their momenta can be chosen as $\mathbf{q}$ and $\mathbf{K}\equiv\mathbf{q}+\mathbf{k}_s$. The relevant loop region has $q, K \sim k_s \ll k_i$. We replace these two propagators by the leading nonlocal terms of their late-time expansions, keeping their polarization tensors, and expand the remaining hard propagators to leading order in the soft momenta. The hard propagators retain their full time dependence and SK ordering. The cut propagators are independent of the SK indices, and their time dependence separates between the two endpoints. The time integrals thus factorize into two tree subgraphs with additional time powers supplied by the cut lines. Multiplying these subgraphs by the soft momentum integral and summing over polarizations gives the nonlocal signal. The branches $\eta=\pm1$ combine into the oscillatory signal retained below. Higher terms in the soft expansion give subleading corrections in the collapsed limit.

Now we will show that the propagators $D^{(T)}$, $D^{(TL)}$, and $D^{(LT)}$ do not contribute to the leading nonlocal signal. On a soft line, each temporal mode function $B^{(T)}$ carries an additional factor of $k\tau$ relative to the spatial modes $B^{(L)}$ and $B^{(\pm)}$. Their nonanalytic contributions are therefore suppressed in the collapsed limit. The contact term in $D^{(T)}$ is analytic in the soft momentum and does not contribute to the nonlocal signal. On a hard line, a temporal endpoint has a vanishing polarization contraction with the spatial components of the attached soft line. A nonzero contraction would require a temporal soft component, which is already suppressed. We therefore omit propagators with temporal components when computing the leading signal below.

All required soft momentum integrals reduce to five dimensionless functions f_r, $r=0,1,2,3,4$. With $z\equiv\hat{\mathbf{q}}\cdot\hat{\mathbf{k}}_s$, we define

$$\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}q^{-2\alpha}K^{-2\beta}z^r = \frac{k_s^{3-2\alpha-2\beta}}{(4\pi)^{3/2}}f_r(\alpha,\beta), \qquad r=0,1,2,3,4. \tag{15}$$

These expressions are understood by analytic continuation from the convergent domain. The first

three functions are

$$f_0(\alpha,\beta) = \frac{\Gamma(\alpha+\beta-3/2)\Gamma(3/2-\alpha)\Gamma(3/2-\beta)}{\Gamma(3-\alpha-\beta)\Gamma(\alpha)\Gamma(\beta)}, \tag{16}$$

$$f_1(\alpha,\beta) = \frac{1}{2}\left[f_0(\alpha+\tfrac{1}{2},\beta-1) - f_0(\alpha-\tfrac{1}{2},\beta) - f_0(\alpha+\tfrac{1}{2},\beta)\right], \tag{17}$$

$$\begin{aligned} f_2(\alpha,\beta) = \frac{1}{4}\big[&f_0(\alpha+1,\beta-2) + f_0(\alpha-1,\beta) + f_0(\alpha+1,\beta) \\ &-2f_0(\alpha,\beta-1) - 2f_0(\alpha+1,\beta-1) + 2f_0(\alpha,\beta)\big]. \end{aligned} \tag{18}$$

The functions f_3 and f_4 are likewise finite sums of f_0 and their explicit expressions are collected in App. A.1.

For the nonlocal branch η, it is convenient to set $x = -\mathrm{i}\eta\tilde{\nu}$ and define

$$\mathcal{J}_r(m,n;x) \equiv f_r\left(x - \frac{m}{2}, x + \frac{n}{2}\right), \qquad r = 0,1,2,3,4. \tag{19}$$

Here m, n are integer shifts of momentum powers, unrelated to m_V. In particular,

$$\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\left(\frac{q}{2}\right)^{-2x}\left(\frac{K}{2}\right)^{-2x} q^m K^{-n} z^r = \frac{4^{2x}k_s^{3-4x+m-n}}{(4\pi)^{3/2}}\mathcal{J}_r(m,n;x). \tag{20}$$

2.2 Bubble Diagram

The bubble contains two quartic vertices, each attached to a pair of external Higgs modes. With the four external momenta understood, the s-channel contribution is

$$\begin{aligned} \langle\delta h^4\rangle'_{\mathrm{Bub}} = \frac{g^4}{8}\sum_{\mathsf{a},\mathsf{b}=\pm1}(-\mathsf{ab})\int_{-\infty}^{0}\frac{\mathrm{d}\tau_1}{|H\tau_1|^2}\int_{-\infty}^{0}\frac{\mathrm{d}\tau_2}{|H\tau_2|^2}G_{\mathsf{a}}(k_1;\tau_1)G_{\mathsf{a}}(k_2;\tau_1)G_{\mathsf{b}}(k_3;\tau_2)G_{\mathsf{b}}(k_4;\tau_2) \\ \times\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}D_{\mathsf{ab},ij}(\mathbf{q};\tau_1,\tau_2)D_{\mathsf{ba},ji}(\mathbf{K};\tau_2,\tau_1), \qquad \mathbf{K} = \mathbf{q}+\mathbf{k}_s. \end{aligned} \tag{21}$$

The symmetry factor $1/2$ which is associated with exchanging the two identical internal lines is included. For the fixed partition (12) | (34), exchanges within either external Higgs pair leave the graph unchanged. There is therefore no additional external-permutation factor in the s-channel signal. And throughout this section, the $W^{\pm}$ contribution is obtained by multiplying the expressions by 2, and the Z contribution by making the replacement $g \to g/c_W$, with the corresponding mass used in each case.

Replacing the two soft propagators by (12)–(13) removes their SK-index dependence, as illustrated in Fig. 2. After summing the polarizations, the oscillatory part of the loop integral is

$$\begin{aligned} &\left[\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}D_{\mathsf{ab},ij}(\mathbf{q};\tau_1,\tau_2)D_{\mathsf{ba},ji}(\mathbf{K};\tau_2,\tau_1)\right]_{\mathrm{NonL}} \\ &\quad\simeq\sum_{\eta=\pm1}(\tau_1\tau_2)^{1+2\mathrm{i}\eta\tilde{\nu}}\frac{\Gamma^4(-\mathrm{i}\eta\tilde{\nu})}{(4\pi)^{7/2}}4^{-2\mathrm{i}\eta\tilde{\nu}}k_s^{3+4\mathrm{i}\eta\tilde{\nu}}\mathcal{I}_{\mathrm{Bub}}(-\mathrm{i}\eta\tilde{\nu}). \end{aligned} \tag{22}$$

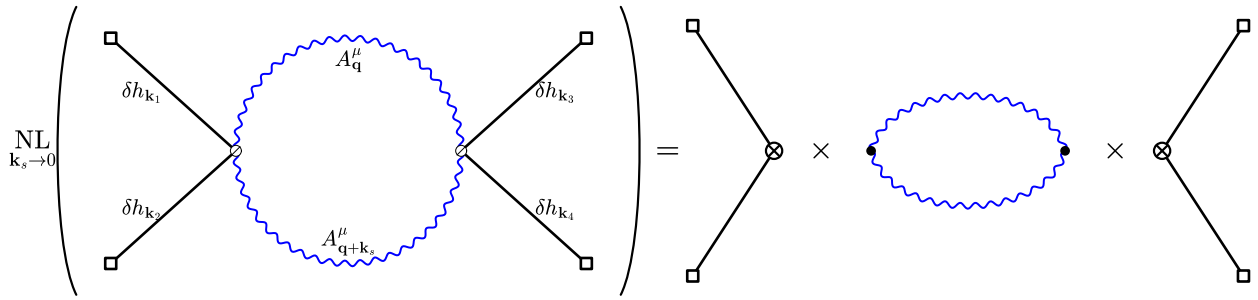


Figure 2: The leading nonlocal part of the bubble in the $k_s \to 0$ limit. The two soft gauge-boson lines supply the central loop factor, while the left and right subgraphs give the two quartic-vertex time integrals.

The dimensionless polarization sum has the compact form

$$\mathcal{I}_{\rm Bub}(x) = \frac{3+2x}{1-2x}\,\mathcal{J}_0(0,0;x) + \frac{16x^2}{(1-2x)^2}\left[\mathcal{J}_0(2,2;x) + 2\mathcal{J}_1(1,2;x) + \mathcal{J}_2(0,2;x)\right], \tag{23}$$

with $x = -\mathrm{i}\eta\tilde{\nu}$ as in (19). The separate polarization contributions are given in App. A.2.

The remaining time integrals factorize. For the left subgraph, at $k_1 \simeq k_2 \simeq k_{12}/2$, we find

$$\sum_{\mathsf{a}=\pm 1} \mathsf{i}\mathsf{a}\int_{-\infty}^{0}\frac{\mathrm{d}\tau}{|H\tau|^2}G_{\mathsf{a}}(k_1;\tau)G_{\mathsf{a}}(k_2;\tau)(-\tau)^{1+2\mathrm{i}\eta\tilde{\nu}} \simeq \frac{H^2 k_{12}^{-2\mathrm{i}\eta\tilde{\nu}}}{4k_1^3k_2^3}\,\mathcal{P}_\eta^{(1)}(\tilde{\nu}), \tag{24}$$

$$\mathcal{P}_\eta^{(1)}(\tilde{\nu}) = \frac{2+\mathrm{i}\eta\tilde{\nu}}{2\mathrm{i}\eta\tilde{\nu}}\mathrm{i}\sinh(\pi\eta\tilde{\nu})\Gamma(2+2\mathrm{i}\eta\tilde{\nu}). \tag{25}$$

The right vertex gives the same expression with $(k_1, k_2, k_{12}) \to (k_3, k_4, k_{34})$. Combining the two vertices with (22), we obtain

$$\left[\langle\delta h^4\rangle'_{\rm Bub}\right]_{\rm NonL} \simeq \frac{g^4H^4k_s^3}{128k_1^3k_2^3k_3^3k_4^3}\left[\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2\mathrm{i}\tilde{\nu}} C^{\rm Bub}(\tilde{\nu}) + \text{c.c.}\right], \tag{26}$$

where the complex signal coefficient is

$$C^{\rm Bub}(\tilde{\nu}) = \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}}\left[\mathcal{P}_+^{(1)}(\tilde{\nu})\right]^2\mathcal{I}_{\rm Bub}(-\mathrm{i}\tilde{\nu}). \tag{27}$$

This definition separates the loop and time integrals from the couplings and diagram symmetry factor. The bracket in (26) equals

$$2|C^{\rm Bub}(\tilde{\nu})|\cos\left[2\tilde{\nu}\log\left(\frac{k_s^2}{4k_{12}k_{34}}\right) + \arg C^{\rm Bub}(\tilde{\nu})\right]. \tag{28}$$

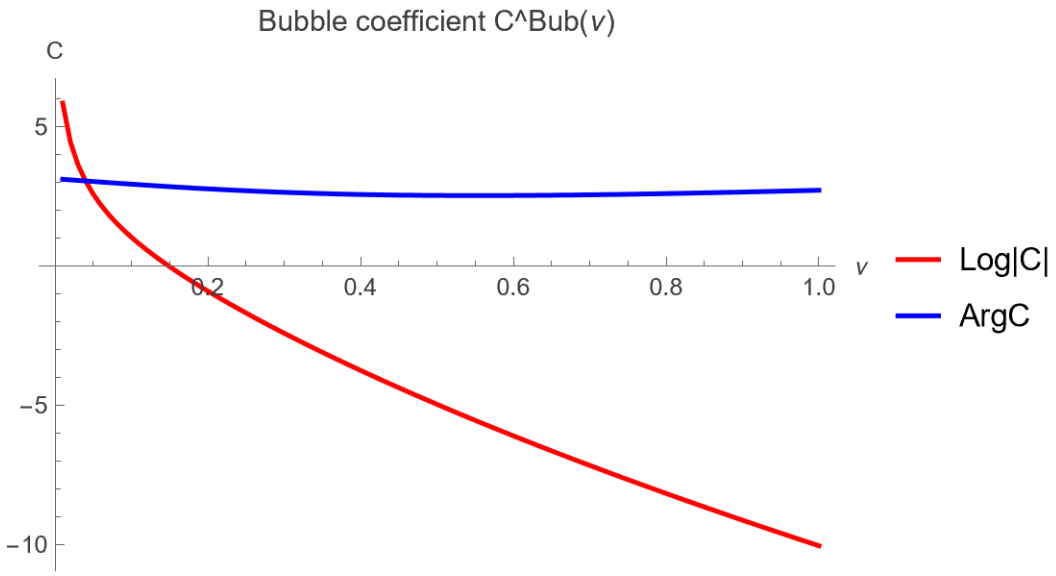


Figure 3: The bubble coefficient defined in (27). The red and blue curves show $\log|C^{\rm Bub}|$ and $\arg C^{\rm Bub}$ in radians, respectively, as functions of $\tilde{\nu}$.

2.3 Triangle Diagram

The triangle contains two cubic vertices and one quartic vertex. We first consider the diagram "Triangle 1". The triangle has a unit diagram symmetry factor. In the conventions of Sec. 2.2, its integral expression is

$$\langle\delta h^4\rangle'_{\rm Tri1} = \frac{g^6h_0^2}{8}\sum_{\mathsf{a},\mathsf{b},\mathsf{c}=\pm1}(-\mathrm{i}\mathsf{abc})\int_{-\infty}^{0}\prod_{v=1}^{3}\frac{\mathrm{d}\tau_v}{|H\tau_v|^2}\,G_{\mathsf{a}}(k_1;\tau_1)G_{\mathsf{b}}(k_2;\tau_2)G_{\mathsf{c}}(k_3;\tau_3)G_{\mathsf{c}}(k_4;\tau_3)$$
$$\times\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}D_{\mathsf{ab},ij}(\mathbf{q}+\mathbf{k}_1;\tau_1,\tau_2)D_{\mathsf{bc},j\ell}(\mathbf{K};\tau_2,\tau_3)D_{\mathsf{ca},\ell i}(\mathbf{q};\tau_3,\tau_1). \tag{29}$$

The nonlocal signal comes from $q, K \sim k_s \ll k_1$. We cut the $\mathbf{q}$ and $\mathbf{K}$ lines and evaluate the remaining hard propagator at $\mathbf{q}+\mathbf{k}_1 \simeq \mathbf{k}_1$. The latter retains its full time dependence and SK ordering. The resulting factorization is shown in Fig. 4.

After summing the soft polarizations, the cut loop takes the form

$$\left[\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}D_{\mathsf{ab},ij}(\mathbf{q}+\mathbf{k}_1;\tau_1,\tau_2)D_{\mathsf{bc},j\ell}(\mathbf{K};\tau_2,\tau_3)D_{\mathsf{ca},\ell i}(\mathbf{q};\tau_3,\tau_1)\right]_{\rm NonL}$$
$$\simeq\sum_{\lambda=L,\pm}\sum_{\eta=\pm1}D^{(\lambda)}_{\mathsf{ab}}(k_1;\tau_1,\tau_2)(\tau_1\tau_2)^{1/2+\mathrm{i}\eta\tilde{\nu}}(-\tau_3)^{1+2\mathrm{i}\eta\tilde{\nu}}\frac{\Gamma^4(-\mathrm{i}\eta\tilde{\nu})}{(4\pi)^{7/2}}4^{-2\mathrm{i}\eta\tilde{\nu}}k_s^{3+4\mathrm{i}\eta\tilde{\nu}}\mathcal{I}^{\lambda}_{\rm Tri}(-\mathrm{i}\eta\tilde{\nu},\theta_1). \tag{30}$$

This equation defines the dimensionless coefficients $\mathcal{I}^{\lambda}_{\rm Tri}(x,\theta)$, with $x=-\mathrm{i}\eta\tilde{\nu}$. After simplification, we surprisingly find that all three coefficients equal $\mathcal{I}_{\rm Bub}(x)/3$ and the apparent angular dependence cancels, as shown in (74).

We next evaluate the left and right subgraphs. The quartic vertex gives the time integral $\mathcal{P}^{(1)}_{\eta}$

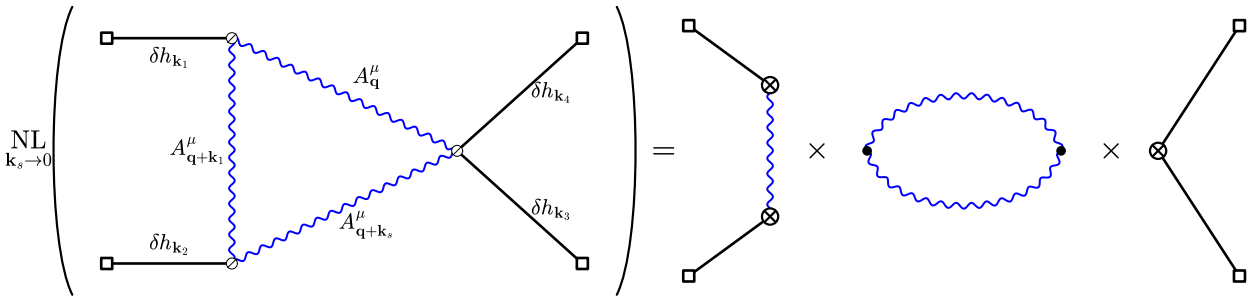


Figure 4: The leading nonlocal part of the first triangle. Cutting the two soft lines leaves a hard gauge-boson exchange on the left and a quartic vertex on the right, joined by the central soft loop factor. The second triangle follows by exchanging the two external pairs.

in (25). The two cubic vertices and the hard line instead define

$$\sum_{\mathsf{a},\mathsf{b}=\pm1}(-\mathsf{ab})\int_{-\infty}^{0}\frac{\mathrm{d}\tau_1\mathrm{d}\tau_2}{|H\tau_1|^2|H\tau_2|^2}G_{\mathsf{a}}(k_1;\tau_1)G_{\mathsf{b}}(k_2;\tau_2)D^{(\lambda)}_{\mathsf{ab}}(k_1;\tau_1,\tau_2)(\tau_1\tau_2)^{1/2+\mathrm{i}\eta\tilde{\nu}}$$
$$\simeq \frac{k_{12}^{-2\mathrm{i}\eta\tilde{\nu}}}{4k_1^3k_2^3}\mathcal{P}^{(2),\lambda}_{\eta}(\tilde{\nu}). \tag{31}$$

At leading order, $k_1 \simeq k_2 \simeq k_{12}/2$. The integral therefore reduces to the double-folded scalar seeds of Ref. [3], which we denote by $\mathcal{M}^{p_1p_2}_{\mathsf{ab}} \equiv \widetilde{\mathcal{I}}^{p_1p_2}_{\mathsf{ab}}(1,1)$. For the transverse modes this follows directly from (7). For the longitudinal mode, the differential relation also produces a contact contribution, and we retain this term together with the scalar-seed contributions. The required seed exponents include $p_0 = -5/2 + \mathrm{i}\eta\tilde{\nu}$, on the boundary of the convergence domain. We define the result by analytic continuation from $p_0 + \epsilon$, $\epsilon > 0$, and take the finite part of the assembled expression at $\epsilon = 0$. The seed normalization, closed forms, and combinations giving $\mathcal{P}^{(2),\lambda}_{\eta}$ are collected in App. A.3.

Finally, the leading signal of the first triangle is

$$\left[\langle\delta h^4\rangle'_{\mathrm{Tri1}}\right]_{\mathrm{NonL}} \simeq \frac{g^6h_0^2H^2k_s^3}{128k_1^3k_2^3k_3^3k_4^3}\left[\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2\mathrm{i}\tilde{\nu}}C^{\mathrm{Tri}}(\tilde{\nu}) + \mathrm{c.c.}\right], \tag{32}$$

where

$$C^{\mathrm{Tri}}(\tilde{\nu}) = \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}}\mathcal{P}^{(1)}_{+}(\tilde{\nu})\sum_{\lambda=L,\pm}\mathcal{P}^{(2),\lambda}_{+}(\tilde{\nu})\mathcal{I}^{\lambda}_{\mathrm{Tri}}(-\mathrm{i}\tilde{\nu},\theta). \tag{33}$$

Since $\mathcal{P}^{(2),+}_{\eta} = \mathcal{P}^{(2),-}_{\eta}$, the two transverse helicities contribute equally. The simplified loop coefficients make C^{Tri} independent of θ at this order and its reduced expression is given in (75).

In the collapsed s channel, the quartic vertex carries either the pair $(3,4)$, as in Triangle 1, or the pair $(1,2)$, as in Triangle 2. For either placement, exchanging the two Higgs legs at the cubic vertices reproduces the same undirected triangle after reflection. Exchanging the two legs at the quartic vertex also leaves the graph unchanged, since both attach to the same vertex. Each placement is therefore counted once. Triangle 2 follows from Triangle 1 by $(1,2) \leftrightarrow (3,4)$, and

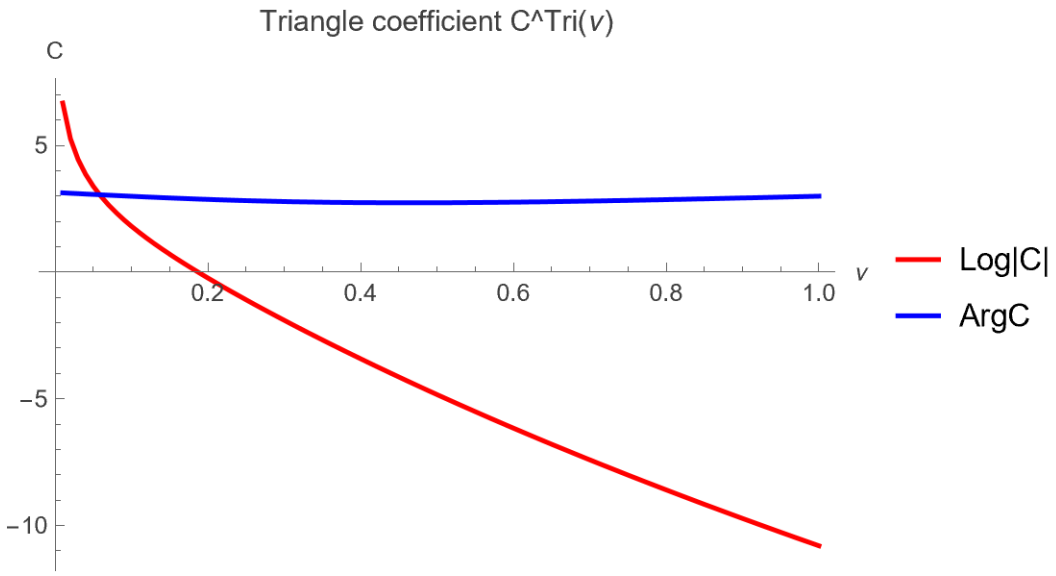


Figure 5: The coefficient $C^{\rm Tri}(\tilde{\nu})$ defined in (33). The red and blue curves show $\log|C^{\rm Tri}|$ and $\arg C^{\rm Tri}$ in radians, respectively.

their leading collapsed signals are equal. Adding these two contributions gives

$$\left[\langle\delta h^4\rangle'_{\rm Tri}\right]_{\rm NonL} \simeq \frac{g^6 h_0^2 H^2 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3}\left\{\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2i\tilde{\nu}}\left[C^{\rm Tri}(\tilde{\nu})+C^{\rm Tri}(\tilde{\nu})\right]+\text{c.c.}\right\}. \tag{34}$$

The nonanalytic momentum dependence is the same as in the bubble, while the hard propagator changes the amplitude and phase. Figure 5 shows the coefficient of a single triangle.

2.4 Box Diagram

We first consider the box with cyclic ordering 1234, shown in Fig. 6. With the external labels fixed, its diagram symmetry factor is unity. The four cubic vertices give the coupling factor $(g^2h_0/2)^4 = g^8h_0^4/16$.

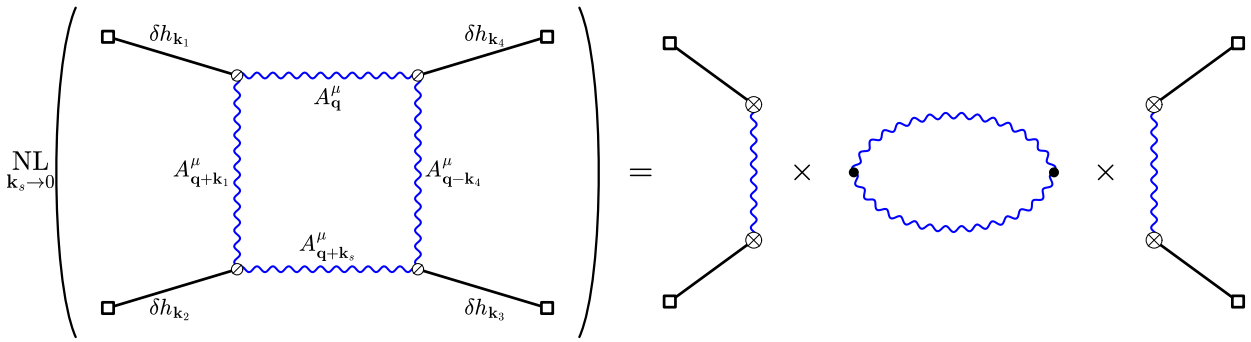


Figure 6: The leading nonlocal part of the box with cyclic ordering 1234. Cutting the two soft lines separates two tree subgraphs, each containing one hard gauge-boson propagator. The central factor contains the nonlocal parts of the two soft propagators and their loop momentum integral.

Choose the loop momenta along the four successive internal lines as $\mathbf{q}+\mathbf{k}_1$, $\mathbf{K} = \mathbf{q}+\mathbf{k}_s$, $\mathbf{q}-\mathbf{k}_4$, and $\mathbf{q}$. In the collapsed limit, the leading nonlocal signal comes from the region $q, K \sim k_s$. Cutting these two soft lines leaves two hard propagators, with momenta approaching $\mathbf{k}_1$ and $\mathbf{k}_3$. Their polarizations are denoted by $\lambda_1, \lambda_3 = L, \pm$. The soft loop gives the dimensionless coefficient $\mathcal{I}^{\lambda_1\lambda_3}_{\rm Box}(-\mathrm{i}\eta\tilde{\nu}, \theta_1, \theta_3, \phi_3)$ defined in (81). Its common momentum factor is $4^{-2\mathrm{i}\eta\tilde{\nu}}k_s^{3+4\mathrm{i}\eta\tilde{\nu}}$, as for the bubble and triangle. As shown in Fig. 6, the time integrals factorize into two copies of (31), one for each hard line. They contribute $\mathcal{P}^{(2),\lambda_1}_\eta(\tilde{\nu})$ and $\mathcal{P}^{(2),\lambda_3}_\eta(\tilde{\nu})$.

The collapsed s channel contains a second inequivalent box, obtained from the diagram shown in Fig. 6 by exchanging the external momenta $\mathbf{k}_1$ and $\mathbf{k}_2$. Exchanging $\mathbf{k}_3$ and $\mathbf{k}_4$ gives the same additional undirected graph, while exchanging both pairs returns the original graph. At leading collapsed order, $\mathbf{k}_2 \simeq -\mathbf{k}_1$, so the first exchange sends (θ_1, ϕ_3) to $(\pi-\theta_1, \phi_3+\pi)$. The coefficient $C^{\rm Box}$ is invariant under this transformation, as follows from its angular decomposition below. The two boxes therefore contribute equally.

For each box, the vertex factor $1/16$ and the normalization $1/4$ from each hard subgraph give $1/256$. Summing the two distinct boxes supplies a further factor 2, giving $1/128$. Including the complex-conjugate soft branch, the total s-channel signal is

$$\left[\langle\delta h^4\rangle'_{\rm Box}\right]_{\rm NonL} \simeq \frac{g^8h_0^4k_s^3}{128k_1^3k_2^3k_3^3k_4^3}\left[\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2\mathrm{i}\tilde{\nu}} \times C^{\rm Box}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) + \text{c.c.}\right], \tag{35}$$

Here $C^{\rm Box}$ denotes the coefficient of the representative box, before summing the external orderings:

$$C^{\rm Box}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}} \sum_{\lambda_1,\lambda_3=L,\pm} \mathcal{P}^{(2),\lambda_1}_+(\tilde{\nu})\mathcal{P}^{(2),\lambda_3}_+(\tilde{\nu}) \times \mathcal{I}^{\lambda_1\lambda_3}_{\rm Box}(-\mathrm{i}\tilde{\nu}, \theta_1, \theta_3, \phi_3). \tag{36}$$

Since the hard transverse propagators have equal time integrals, the polarization sum reduces to the LL, LH, and HH groups in (82) where H means $+$ component contributions plus $-$ component contributions. Their explicit expressions are given in App. A.4.

The result contains five angular structures:

$$\begin{aligned} C^{\mathrm{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = {} & A_{1,g}(\tilde{\nu}) + A_{2,g}(\tilde{\nu})(\cos^2\theta_1 + \cos^2\theta_3) + A_{3,g}(\tilde{\nu})\cos^2\theta_1\cos^2\theta_3 \\ & + B_g(\tilde{\nu})\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3 \\ & + D_g(\tilde{\nu})\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3). \end{aligned} \tag{37}$$

The five complex functions depend only on $\tilde{\nu}$. The signal is symmetric under $\theta_1 \leftrightarrow \theta_3$ and $\phi_3 \to -\phi_3$. Its oscillatory momentum dependence is the same as in (28), with an angle-dependent amplitude and phase supplied by C^{Box}.

Figures 7 and 8 show the five coefficient functions. Their magnitudes decrease over the displayed mass range. The jumps in the plotted phases reflect the principal branch of arg, defined modulo 2π.

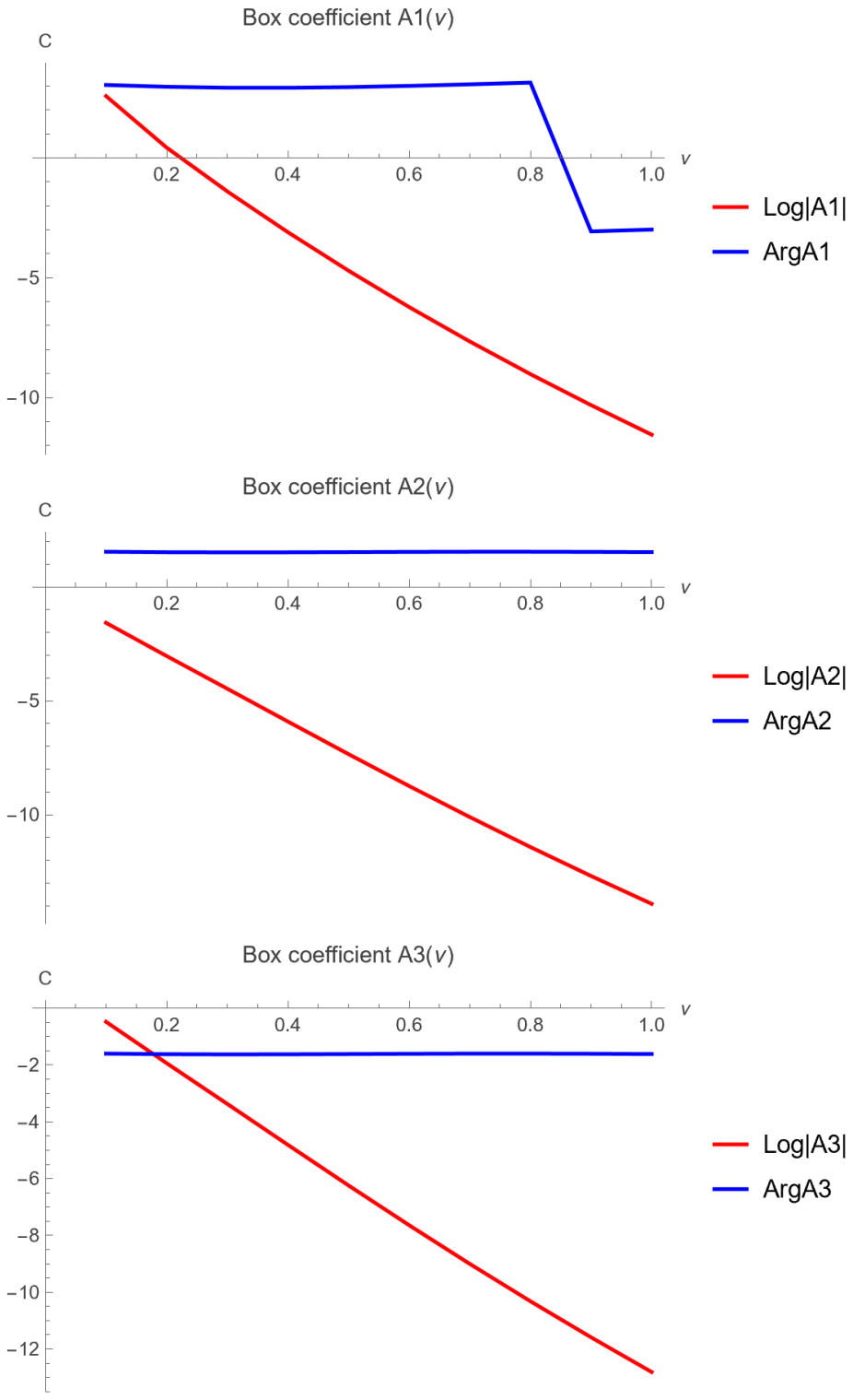


Figure 7: The angular coefficients $A_{1,g}$, $A_{2,g}$, and $A_{3,g}$ (top to bottom) in (37) for the box diagram. Red curves show the logarithmic magnitudes and blue curves show the principal phases in radians.

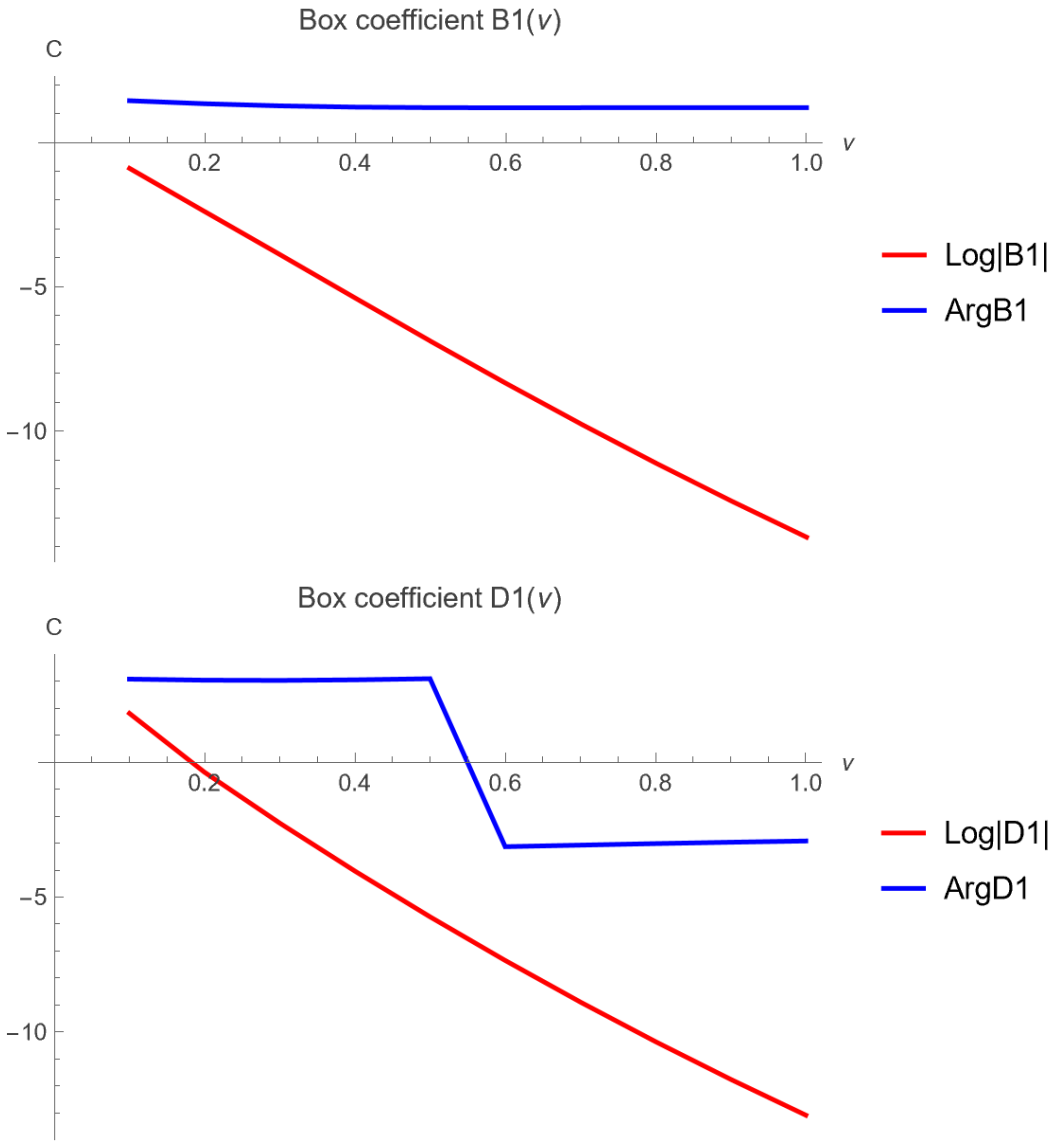


Figure 8: The angular coefficients B_g (top, labelled $B1$) and D_g (bottom, labelled $D1$) in (37). The plotting conventions are the same as in Fig. 7.

3 Top Quark Signal

3.1 Preliminaries

We next compute the Higgs four-point function generated by the top-quark loop. We use the collapsed kinematics of (1) and the external propagator (4). The top mass is treated as constant and the chemical potential vanishes. The Yukawa coupling and mass parameter are

$$g_t \equiv \frac{y_t}{\sqrt{2}}, \qquad m_t = g_t h_0, \qquad \tilde{\nu} \equiv \frac{m_t}{H} > 0. \tag{38}$$

Thus the relation between $\tilde{\nu}$ and the mass differs from that of the vector field.

Let t and t^c denote the conformally rescaled left-handed top and the left-handed charge conjugate of the right-handed top. Rescaling the covariant fields by $a^{3/2}$, with $a = -1/(H\tau)$, gives

$$\mathcal{L} = \mathrm{i}t^\dagger \bar{\sigma}^\mu \partial_\mu t + \mathrm{i}t^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(m_t + g_t \delta h)(t t^c + \mathrm{h.c.}). \tag{39}$$

Here t^c represents the right-handed degree of freedom in left-handed notation. Following the two-component description of Ref. [5], we set $t = (\psi_1 + \mathrm{i}\psi_2)/\sqrt{2}$ and $t^c = (\psi_1 - \mathrm{i}\psi_2)/\sqrt{2}$. The action becomes

$$\mathcal{L} = \sum_{I=1}^{2} \left[\mathrm{i}\psi_I^\dagger \bar{\sigma}^\mu \partial_\mu \psi_I - \frac{a}{2}(m_t + g_t \delta h)(\psi_I \psi_I + \mathrm{h.c.}) \right]. \tag{40}$$

The two sectors are identical. Each vertex retains one factor of $a(\tau)$ and the propagators below are those of the rescaled field and contain no factors of $a^{-3/2}$.

For one sector, the normalized Bunch–Davies state mode functions can be written as

$$u_+ = v_- = \frac{\tilde{\nu}}{\sqrt{2x}} W_{-1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}x), \qquad u_- = v_+ = \frac{\mathrm{i}}{\sqrt{2x}} W_{1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}x), \qquad x = -k\tau. \tag{41}$$

The helicity is $h = \pm 1$, $|u_s|^2 + |v_s|^2 = 1$, and the creation part of the field contains v_h^*. We use the three propagators [6]

$$\begin{aligned} D_{\mathsf{ab},\alpha\dot{\beta}}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi_{\mathsf{a},\alpha}(\mathbf{k},\tau_1) \psi^\dagger_{\mathsf{b},\dot{\beta}}(-\mathbf{k},\tau_2)\rangle', \\ \widehat{D}_{\mathsf{ab},\alpha\beta}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi_{\mathsf{a},\alpha}(\mathbf{k},\tau_1) \psi_{\mathsf{b},\beta}(-\mathbf{k},\tau_2)\rangle', \\ \check{D}_{\mathsf{ab},\dot{\alpha}\dot{\beta}}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi^\dagger_{\mathsf{a},\dot{\alpha}}(\mathbf{k},\tau_1) \psi^\dagger_{\mathsf{b},\dot{\beta}}(-\mathbf{k},\tau_2)\rangle'. \end{aligned} \tag{42}$$

Here T_C denotes ordering along the Schwinger–Keldysh closed time contour C. Their mode sums and fermionic reversal rules are given in App. B.1. For each of the three types, the time-ordered components obey

$$\mathcal{D}_{++} = \theta_{12}\mathcal{D}_{-+} + \theta_{21}\mathcal{D}_{+-}, \qquad \mathcal{D}_{--} = \theta_{21}\mathcal{D}_{-+} + \theta_{12}\mathcal{D}_{+-}, \qquad \theta_{ij} = \theta(\tau_i - \tau_j). \tag{43}$$

To display the soft expansion, write $\sigma_p = \boldsymbol{\sigma}\cdot\hat{\mathbf{p}}$ and $\boldsymbol{\epsilon} = \mathrm{i}\sigma^2$, with $\epsilon_{12} = 1$ and $\epsilon^{12} = -1$. Define the common branch coefficient

$$c_\eta = \frac{2^{-1-2\mathrm{i}\eta\tilde{\nu}}}{\pi}\Gamma^2(\tfrac{1}{2} - \mathrm{i}\eta\tilde{\nu}), \qquad c_- = c_+^*, \qquad |c_+| = \frac{1}{2\cosh(\pi\tilde{\nu})}. \tag{44}$$

At $|p\tau_1|, |p\tau_2| \ll 1$, the required nonlocal terms are

$$
\begin{aligned}
[D_{\mathsf{ab}}]_{\mathrm{NonL}} &= -\frac{1}{2}\sum_{\eta=\pm1} c_\eta (p^2\tau_1\tau_2)^{\mathrm{i}\eta\tilde{\nu}} \left[\sigma_p + \frac{\mathrm{i}p(\tau_1-\tau_2)}{1+2\mathrm{i}\eta\tilde{\nu}}\mathbf{1}_2 + \cdots\right], \\
[\widehat{D}_{\mathsf{ab}}]_{\mathrm{NonL}} &= -\frac{1}{2}\sum_{\eta=\pm1} \eta c_\eta (p^2\tau_1\tau_2)^{\mathrm{i}\eta\tilde{\nu}} \left[\sigma_p + \frac{\mathrm{i}p(\tau_1+\tau_2)}{1+2\mathrm{i}\eta\tilde{\nu}}\mathbf{1}_2 + \cdots\right]\boldsymbol{\epsilon}, \\
[\check{D}_{\mathsf{ab}}]_{\mathrm{NonL}} &= -\frac{1}{2}\sum_{\eta=\pm1} \eta c_\eta (p^2\tau_1\tau_2)^{\mathrm{i}\eta\tilde{\nu}} \boldsymbol{\epsilon}\left[\sigma_p - \frac{\mathrm{i}p(\tau_1+\tau_2)}{1+2\mathrm{i}\eta\tilde{\nu}}\mathbf{1}_2 + \cdots\right].
\end{aligned} \tag{45}
$$

The omitted terms are second order in the soft momentum at fixed times. These nonlocal expressions are independent of the SK indices. They allow the two soft lines to be cut, while the remaining hard lines retain their full time dependence and SK ordering [4, 6].

The Yukawa interaction gives two vertex types, labelled $r = \mathrm{L}, \overline{\mathrm{L}}$ for $\psi\psi$ and $\psi^\dagger\psi^\dagger$, respectively. Their amputated rules are $-\mathsf{i}\mathsf{a}g_t a(\tau) J_r$, where $J_{\mathrm{L}} = \boldsymbol{\epsilon}$ and $J_{\overline{\mathrm{L}}} = -\boldsymbol{\epsilon}$.

3.2 The Box Diagram and Soft Expansion

For a fixed ordering 1234, including both sectors and all colors, the box is

$$
\begin{aligned}
\langle \delta h^4 \rangle'_{t,1234} = &-N_c g_t^4 \sum_{\mathsf{a}_1,\ldots,\mathsf{a}_4=\pm1} \left(\prod_{v=1}^4 \mathsf{a}_v\right) \int_{-\infty}^0 \prod_{v=1}^4 [\mathrm{d}\tau_v\, a(\tau_v) G_{\mathsf{a}_v}(k_v;\tau_v)] \\
&\times \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \sum_{r_1,\ldots,r_4} \mathrm{tr}_2[D_{12}J_{r_2}D_{23}J_{r_3}D_{34}J_{r_4}D_{41}J_{r_1}],
\end{aligned} \tag{46}
$$

where $D_{ij} = D^{r_i r_j}_{\mathsf{a}_i\mathsf{a}_j}(\mathbf{p}_{ij};\tau_i,\tau_j)$ and tr_2 contracts the two spinor components. The minus sign is the closed-fermion-loop sign. A single Majorana sector has the coefficient $-g_t^4/2$ in this directed-cycle convention; multiplication by $2N_c$ gives the prefactor in (46). The vertex-type sum does not include external permutations.

Choose the successive momenta as $\mathbf{q}+\mathbf{k}_1$, $\mathbf{K} = \mathbf{q}+\mathbf{k}_s$, $\mathbf{q}-\mathbf{k}_4$, and $\mathbf{q}$. In the double-soft region $q, K \sim k_s$, it is useful to introduce centered hard momenta,

$$
\mathbf{k}_L = \frac{\mathbf{k}_1-\mathbf{k}_2}{2}, \qquad \mathbf{k}_R = \frac{\mathbf{k}_3-\mathbf{k}_4}{2}, \qquad \boldsymbol{\ell} = \mathbf{q}+\frac{\mathbf{k}_s}{2}. \tag{47}
$$

The hard lines then carry $\mathbf{k}_L+\boldsymbol{\ell}$ and $\mathbf{k}_R+\boldsymbol{\ell}$. In particular the right momentum is exactly $\mathbf{q}-\mathbf{k}_4 = \mathbf{q}+\mathbf{k}_s+\mathbf{k}_3$. At leading collapsed order, $k_L \simeq k_{12}/2$ and $k_R \simeq k_{34}/2$.

For the same-sign soft branch $\eta = +1$, attaching the soft endpoints to each hard line gives matrices V_{12} and V_{34}. Their normalization is fixed in App. B.2 so that the cut spin factor of this ordering is

$$
c_+^2 (qK)^{2\mathrm{i}\tilde{\nu}} \prod_{v=1}^4 (-\tau_v)^{\mathrm{i}\tilde{\nu}}\, \mathrm{tr}_2[V_{12}\sigma_K V_{34}\sigma_q]. \tag{48}
$$

The four orderings contributing to the collapsed s channel are $1234, 1243, 1342, 1432$. They replace the trace by

$$
\mathrm{tr}_2[(V_{12}+V_{21})\sigma_K(V_{34}+V_{43})\sigma_q]. \tag{49}
$$

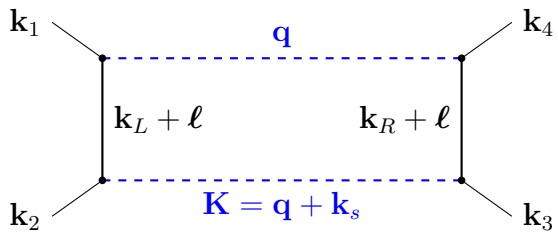


Figure 9: The double-soft region of the top-quark box. Cutting the two dashed lines separates the left and right hard exchanges. All momentum labels are vectors; the hard propagators retain their complete time dependence.

Every time and SK label remains attached to its physical external vertex, so all four terms have the same exact Higgs factors.

At zeroth soft order, the vertex-type sum at a hard end is proportional to its direction. Writing its scalar coefficient as $F_{0;\mathsf{ab}}$, one finds

$$\begin{aligned} V_{12}^{(0)} &= -F_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\sigma}\cdot\mathbf{n}_L, & V_{21}^{(0)} &= +F_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\sigma}\cdot\mathbf{n}_L, \\ V_{12}^{(0)} + V_{21}^{(0)} &= 0, & V_{34}^{(0)} + V_{43}^{(0)} &= 0, \end{aligned} \tag{50}$$

where $\mathbf{n}_L = \widehat{\mathbf{k}}_L$ and $\mathbf{n}_R = \widehat{\mathbf{k}}_R$. The scalar coefficient is invariant under reversal of its two time and SK labels. Thus the cancellation holds before the time integrations or the SK sum. A single ordering has a nonzero zeroth-order trace, but the sum removes the candidate $k_s^{3\pm 4i\tilde{\nu}}$ oscillatory signal.

The first surviving term has one soft power at each hard end. Both the magnitude and direction of the hard momentum must be expanded, together with the first-order soft terms in (45). Their sum can be organized as

$$\begin{aligned} V_{12}^{(1)} + V_{21}^{(1)} &= -\frac{2}{k_L}\boldsymbol{\sigma}\cdot\mathbf{v}_L, & \mathbf{v}_L &= \Pi_1^L\boldsymbol{\ell} + \Pi_2^L\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell}), \\ V_{34}^{(1)} + V_{43}^{(1)} &= -\frac{2}{k_R}\boldsymbol{\sigma}\cdot\mathbf{v}_R, & \mathbf{v}_R &= \Pi_1^R\boldsymbol{\ell} + \Pi_2^R\mathbf{n}_R(\mathbf{n}_R\cdot\boldsymbol{\ell}). \end{aligned} \tag{51}$$

The two scalar functions Π_i depend on the endpoint times and SK labels but not on $\mathbf{q}$; their explicit expressions are given in (106). Terms with a zeroth-order end, including the second-order expansion at the other end, vanish by (50). Consequently the first nonzero oscillatory signal scales as $k_s^{5\pm 4i\tilde{\nu}}$.

3.3 Factorized Loop and Time Integrals

The Pauli trace in (49) reduces the remaining loop to a scalar integral. We define its dimensionless coefficients by

$$\begin{aligned} &\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}(qK)^{2i\tilde{\nu}-1}\left[2(\mathbf{v}_L\cdot\boldsymbol{\ell})(\mathbf{v}_R\cdot\boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L\cdot\mathbf{k}_s)(\mathbf{v}_R\cdot\mathbf{k}_s) - (\mathbf{v}_L\cdot\mathbf{v}_R)(\ell^2 - k_s^2/4)\right] \\ &\qquad = k_s^{5+4i\tilde{\nu}}\sum_{i,j=1}^{2}\mathcal{I}_{t,\mathrm{Box}}^{ij}(\tilde{\nu},\theta_1,\theta_3,\phi_3)\Pi_i^L\Pi_j^R. \end{aligned} \tag{52}$$

The trace supplies a further factor $8/(k_L k_R)$, consisting of 2 from the Pauli identity and 4 from the two ordering sums. At this order the angles of the centered hard directions can be identified with those in Sec. 2.1. Every polynomial momentum moment in (52) reduces to the scalar convolution f_0 of (16). The resulting coefficients satisfy

$$\mathcal{I}^{11}_{t,\mathrm{Box}} = 3\mathcal{I}^{12}_{t,\mathrm{Box}} = 3\mathcal{I}^{21}_{t,\mathrm{Box}}. \tag{53}$$

These three coefficients are independent of angles. All angular dependence is carried by $\mathcal{I}^{22}_{t,\mathrm{Box}}$, whose explicit expression is collected in App. B.2. The integrals are defined by analytic continuation, as in (15); polynomial contact terms are excluded.

The two time integrations on each side are now independent. For their leading collapsed values, set the two external energies equal to the hard magnitude k and define

$$\begin{aligned}\frac{4k^6}{H^4}\sum_{\mathsf{a},\mathsf{b}=\pm1}\mathsf{ab}\int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\,(-\tau_1)^{\mathrm{i}\tilde{\nu}-1}(-\tau_2)^{\mathrm{i}\tilde{\nu}-1}G_{\mathsf{a}}(k;\tau_1)G_{\mathsf{b}}(k;\tau_2)\Pi_{i;\mathsf{ab}}(k;\tau_1,\tau_2)\\ = k^{-2\mathrm{i}\tilde{\nu}}\mathcal{P}_{t,i}(\tilde{\nu}), \qquad i=1,2.\end{aligned} \tag{54}$$

The left and right subgraphs use $k = k_L$ and $k = k_R$, respectively. The coefficient $\mathcal{P}_{t,1}$ includes the hard-direction correction and the two soft-endpoint corrections; $\mathcal{P}_{t,2}$ contains the hard-magnitude derivative and its associated direction correction. The exact external propagators were common factors throughout the cancellation and loop integral. Corrections to their equal-energy limit enter beyond the leading nonzero term.

These time integrals reduce to the Whittaker seed of Ref. [3], denoted here by $\mathcal{K}^{\kappa;p_1p_2}_{\mathsf{ab}}$ with $\kappa = \pm 1/2$. This plays the same role for $\mathcal{P}_{t,i}$ as $\mathcal{M}^{p_1p_2}_{\mathsf{ab}}$ does for the gauge-boson coefficients. Its scalar kernel has a different normalization from $D^{(\mathrm{s})}$ in (6). The continuation in κ is an auxiliary method of evaluating the fermion modes; the physical top chemical potential remains zero. The seed normalization, its closed form and the combinations giving $\mathcal{P}_{t,i}$ are given in App. B.3. A common endpoint regulator is removed after the complete SK sum has been assembled.

3.4 Oscillatory Signal

Combining the soft loop and the two hard subgraphs, and adding the complex-conjugate branch, gives

$$\begin{aligned}\left[\langle\delta h^4\rangle'_{t,s}\right]_{\mathrm{osc}} \simeq\; &-\frac{N_c g_t^4 H^4 k_s^5}{16k_1^3k_2^3k_3^3k_4^3k_Lk_R}\\ &\times\left[\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2\mathrm{i}\tilde{\nu}} C_t^{\mathrm{Box}}(\tilde{\nu},\theta_1,\theta_3,\phi_3)+\mathrm{c.c.}\right],\end{aligned} \tag{55}$$

where

$$C_t^{\mathrm{Box}} = 8c_+^2\,16^{2\mathrm{i}\tilde{\nu}}\sum_{i,j=1}^{2}\mathcal{P}_{t,i}\mathcal{P}_{t,j}\mathcal{I}^{ij}_{t,\mathrm{Box}}. \tag{56}$$

All four s-channel orderings are included. The phase factor $16^{2\mathrm{i}\tilde{\nu}}$ converts the natural hard scale k_Lk_R to the momentum ratio used in Sec. 2, since $4k_{12}k_{34} \simeq 16k_Lk_R$.

The angular decomposition takes the same form as (37):

$$C_t^{\rm Box} = A_{1,t} + A_{2,t}(\cos^2\theta_1 + \cos^2\theta_3) + A_{3,t}\cos^2\theta_1\cos^2\theta_3 + B_t\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3 + D_t\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3). \tag{57}$$

The five coefficients depend only on $\tilde{\nu}$ and satisfy $A_{3,t} = -3A_{2,t}$. Their analytic expressions are given in App. B.4. In particular all nonconstant angular terms are proportional to $\mathcal{P}_{t,2}^2$. The result is even under either Higgs-pair exchange and under parity.

At fixed angles the oscillatory bracket in (55) is

$$2|C_t^{\rm Box}|\cos\left[2\tilde{\nu}\log\left(\frac{k_s^2}{4k_{12}k_{34}}\right) + \arg C_t^{\rm Box}\right]. \tag{58}$$

The frequency in $\log k_s$ is $4m_t/H$. Compared with the gauge-boson signals, the top oscillation carries an additional soft suppression $k_s^2/(k_L k_R)$ from (50). The coefficient determines its mass-dependent and angle-dependent amplitude and phase.

3.5 Nonoscillatory Signal

Opposite branches on the two cut lines produce a nonoscillatory contribution. Their zeroth-order hard matrices are proportional to $\mathbf{1}_2$, with scalar coefficient F_1 defined in App. B.1. They are even under the external exchange and therefore survive the ordering sum. The spin trace is proportional to $2\hat{\mathbf{q}}\cdot\widehat{\mathbf{K}}$. The corresponding soft convolution is

$$\int\frac{{\rm d}^3\mathbf{q}}{(2\pi)^3}(q^2)^{-{\rm i}\tilde{\nu}}(K^2)^{{\rm i}\tilde{\nu}}\hat{\mathbf{q}}\cdot\widehat{\mathbf{K}} = \frac{k_s^3}{(4\pi)^{3/2}}\frac{2\sqrt{\pi}}{3}\tilde{\nu}(1+\tilde{\nu}^2)\coth(\pi\tilde{\nu}). \tag{59}$$

Both hard subgraphs give the same real time coefficient $\mathcal{P}_{t,\rm mix}$, evaluated through $\mathcal{K}$ in (119). Summing the two opposite-branch assignments and the four orderings yields

$$\left[\langle\delta h^4\rangle'_{t,s}\right]_{\rm mix} \simeq \frac{N_c g_t^4 H^4 k_s^3}{16k_1^3k_2^3k_3^3k_4^3}\frac{2\tilde{\nu}(1+\tilde{\nu}^2)}{3\pi\sinh(2\pi\tilde{\nu})}\mathcal{P}_{t,\rm mix}^2. \tag{60}$$

This term has no angular dependence at leading order. Although it does not oscillate, $k_s^3 = (\mathbf{k}_s^2)^{3/2}$ is nonanalytic in the soft momentum components and is not a polynomial contact contribution.

Equations (55) and (60) give the leading oscillatory and nonoscillatory nonanalytic terms, respectively, for fixed $\tilde{\nu} > 0$ in one collapsed channel. The coincident branches at $\tilde{\nu} = 0$ require a combined limit. Analytic contact terms and the conversion of Higgs fluctuations into curvature perturbations are not included.

4 Conclusions

A Gauge Boson Integrals and Coefficients

A.1 Loop Integral Formulae

The identity $z = (K^2 - q^2 - k_s^2)/(2qk_s)$ reduces all angular moments in (15) to the scalar integral f_0:

$$f_r(\alpha,\beta) = \frac{1}{2^r}\sum_{u+v+w=r}\frac{r!}{u!v!w!}(-1)^{v+w} f_0\left(\alpha+\frac{r}{2}-v,\beta-u\right), \tag{61}$$

where the sum runs over nonnegative integers. For the third and fourth moments this gives

$$\begin{aligned} f_3(\alpha,\beta) = \frac{1}{8}\Big[&f_0(\alpha+\tfrac{3}{2},\beta-3) - 3f_0(\alpha+\tfrac{3}{2},\beta-2) - 3f_0(\alpha+\tfrac{1}{2},\beta-2)\\ &+ 3f_0(\alpha+\tfrac{3}{2},\beta-1) + 6f_0(\alpha+\tfrac{1}{2},\beta-1) + 3f_0(\alpha-\tfrac{1}{2},\beta-1)\\ &- f_0(\alpha+\tfrac{3}{2},\beta) - 3f_0(\alpha+\tfrac{1}{2},\beta) - 3f_0(\alpha-\tfrac{1}{2},\beta) - f_0(\alpha-\tfrac{3}{2},\beta)\Big], \end{aligned} \tag{62}$$

$$\begin{aligned} f_4(\alpha,\beta) = \frac{1}{16}\Big[&f_0(\alpha+2,\beta-4) - 4f_0(\alpha+2,\beta-3) - 4f_0(\alpha+1,\beta-3)\\ &+ 6f_0(\alpha+2,\beta-2) + 12f_0(\alpha+1,\beta-2) + 6f_0(\alpha,\beta-2)\\ &- 4f_0(\alpha+2,\beta-1) - 12f_0(\alpha+1,\beta-1) - 12f_0(\alpha,\beta-1)\\ &- 4f_0(\alpha-1,\beta-1) + f_0(\alpha+2,\beta) + 4f_0(\alpha+1,\beta)\\ &+ 6f_0(\alpha,\beta) + 4f_0(\alpha-1,\beta) + f_0(\alpha-2,\beta)\Big]. \end{aligned} \tag{63}$$

A.2 Bubble Coefficients

For $x = -\mathrm{i}\eta\tilde{\nu}$, the relative weight of a longitudinal nonlocal propagator is

$$\beta(x) \equiv \frac{(x+1/2)^2}{1/4-x^2} = \frac{1+2x}{1-2x}. \tag{64}$$

Define $\mathcal{I}_{\sigma_1\sigma_2}(x)$ by restricting the loop in (22) to the polarizations (σ_1,σ_2) and replacing $\mathcal{I}_{\mathrm{Bub}}$ on its right-hand side by $\mathcal{I}_{\sigma_1\sigma_2}$. The independent coefficients are

$$\mathcal{I}_{LL}(x) = [\beta(x)]^2\left[\mathcal{J}_0(2,2;x) + 2\mathcal{J}_1(1,2;x) + \mathcal{J}_2(0,2;x)\right], \tag{65}$$

$$\mathcal{I}_{L+}(x) = \frac{\beta(x)}{2}\left[\mathcal{J}_0(0,2;x) - \mathcal{J}_2(0,2;x)\right], \tag{66}$$

$$\begin{aligned} \mathcal{I}_{++}(x) = \frac{1}{4}\big[&\mathcal{J}_0(0,0;x) + \mathcal{J}_0(2,2;x) + \mathcal{J}_2(0,2;x)\\ &+2\mathcal{J}_0(1,1;x) + 2\mathcal{J}_1(1,2;x) + 2\mathcal{J}_1(0,1;x)\big], \end{aligned} \tag{67}$$

$$\begin{aligned} \mathcal{I}_{+-}(x) = \frac{1}{4}\big[&\mathcal{J}_0(0,0;x) + \mathcal{J}_0(2,2;x) + \mathcal{J}_2(0,2;x)\\ &-2\mathcal{J}_0(1,1;x) + 2\mathcal{J}_1(1,2;x) - 2\mathcal{J}_1(0,1;x)\big]. \end{aligned} \tag{68}$$

The remaining entries satisfy $\mathcal{I}_{--} = \mathcal{I}_{++}$, $\mathcal{I}_{-+} = \mathcal{I}_{+-}$, and $\mathcal{I}_{L-} = \mathcal{I}_{+L} = \mathcal{I}_{-L} = \mathcal{I}_{L+}$. Consequently,

$$\mathcal{I}_{\text{Bub}} = \mathcal{I}_{LL} + 4\mathcal{I}_{L+} + 2\mathcal{I}_{++} + 2\mathcal{I}_{+-}, \tag{69}$$

which reduces to (23) using the moment relations in App. A.1.

For the $\eta = +1$ branch, the fully reduced sum is

$$\mathcal{I}_{\text{Bub}}(-\mathrm{i}\tilde{\nu}) = \frac{3 \cdot 2^{-4-4\mathrm{i}\tilde{\nu}}(3\mathrm{i} - 2\tilde{\nu})^2\Gamma(-3/2 - 2\mathrm{i}\tilde{\nu})\Gamma(3 + 2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi[1 + (3\mathrm{i} - 2\tilde{\nu})\tilde{\nu}]^2}. \tag{70}$$

Substituting this into (27) gives

$$\begin{aligned} C^{\text{Bub}}(\tilde{\nu}) = \frac{3(\tilde{\nu} - \mathrm{i})(\tilde{\nu} - 2\mathrm{i})^4}{256\pi^{7/2}\tilde{\nu}^2(2\tilde{\nu} - \mathrm{i})}\Gamma^2(-2 - \mathrm{i}\tilde{\nu})\Gamma^2(5/2 + \mathrm{i}\tilde{\nu}) \\ \times \Gamma(-3/2 - 2\mathrm{i}\tilde{\nu})\Gamma(1 + 2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu}). \end{aligned} \tag{71}$$

For real $\tilde{\nu} > 0$, the opposite nonlocal branch is their complex conjugate.

A.3 Triangle Coefficients

A.3.1 Polarization sum

Let $x = -\mathrm{i}\eta\tilde{\nu}$ and let $\beta(x)$ be the longitudinal weight in (64). Summing the nine soft-polarization combinations gives

$$\begin{aligned} \mathcal{I}^L_{\text{Tri}}(x,\theta) = [\beta(x)]^2 \Big\{ &\frac{\sin^2\theta}{2}\left[\mathcal{J}_0(2,2;x) + \mathcal{J}_1(1,2;x) - \mathcal{J}_2(2,2;x) - \mathcal{J}_3(1,2;x)\right] \\ &+ \cos^2\theta\left[\mathcal{J}_2(2,2;x) + \mathcal{J}_3(1,2;x) + \mathcal{J}_1(1,2;x) + \mathcal{J}_2(0,2;x)\right]\Big\} \\ &+ \frac{\beta(x)(1+\cos^2\theta)}{2}\left[\mathcal{J}_0(0,2;x) - \mathcal{J}_2(0,2;x)\right] \\ &+ \frac{\sin^2\theta}{2}\left[\mathcal{J}_2(2,2;x) + \mathcal{J}_3(1,2;x) + \mathcal{J}_1(1,2;x) + \mathcal{J}_2(0,2;x) + \mathcal{J}_0(0,0;x)\right] \\ &+ \cos^2\theta\left[\mathcal{J}_0(2,2;x) - \mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) - \mathcal{J}_3(1,2;x)\right]. \end{aligned} \tag{72}$$

The terms proportional to β^2, β, and 1 arise from two longitudinal, mixed, and two transverse soft lines, respectively.

For a hard line with helicity $+$, the corresponding expression is

$$\begin{aligned} \mathcal{I}^+_{\text{Tri}}(x,\theta) = [\beta(x)]^2 \Big\{ &\frac{1+\cos^2\theta}{4}\left[\mathcal{J}_0(2,2;x) - \mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) - \mathcal{J}_3(1,2;x)\right] \\ &+ \frac{\sin^2\theta}{2}\left[\mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) + \mathcal{J}_3(1,2;x) + \mathcal{J}_2(0,2;x)\right]\Big\} \\ &+ \frac{\beta(x)(2+\sin^2\theta)}{4}\left[\mathcal{J}_0(0,2;x) - \mathcal{J}_2(0,2;x)\right] \\ &+ \frac{1+\cos^2\theta}{4}\left[\mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) + \mathcal{J}_3(1,2;x) + \mathcal{J}_2(0,2;x) + \mathcal{J}_0(0,0;x)\right] \\ &+ \frac{\sin^2\theta}{2}\left[\mathcal{J}_0(2,2;x) - \mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) - \mathcal{J}_3(1,2;x)\right]. \end{aligned} \tag{73}$$

At zero chemical potential, reducing the Gamma functions gives the same result for all three hard-line polarizations:

$$\begin{aligned}\mathcal{I}^{L}_{\rm Tri}(-\mathrm{i}\tilde{\nu},\theta) &= \mathcal{I}^{+}_{\rm Tri}(-\mathrm{i}\tilde{\nu},\theta) = \mathcal{I}^{-}_{\rm Tri}(-\mathrm{i}\tilde{\nu},\theta) = \frac{1}{3}\mathcal{I}_{\rm Bub}(-\mathrm{i}\tilde{\nu}) \\ &= \frac{2^{-4-4\mathrm{i}\tilde{\nu}}(3\mathrm{i}-2\tilde{\nu})^2\Gamma(-3/2-2\mathrm{i}\tilde{\nu})\Gamma(3+2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi[1+(3\mathrm{i}-2\tilde{\nu})\tilde{\nu}]^2}. \end{aligned} \tag{74}$$

Summing over the hard-line polarizations reproduces $\mathcal{I}_{\rm Bub}$ by polarization completeness. Thus the angular dependence of the separate soft-polarization contributions cancels in their sum. The coefficient of a single triangle reduces to

$$C^{\rm Tri}(\tilde{\nu}) = \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{3(4\pi)^{7/2}}\mathcal{P}^{(1)}_{+}(\tilde{\nu})\left[\mathcal{P}^{(2),L}_{+}(\tilde{\nu}) + 2\mathcal{P}^{(2),+}_{+}(\tilde{\nu})\right]\mathcal{I}_{\rm Bub}(-\mathrm{i}\tilde{\nu}). \tag{75}$$

A.3.2 Time-integral coefficients

For the scalar propagator of (6), the double-folded seed is defined by

$$\mathcal{M}^{p_1p_2}_{\mathsf{ab}} \equiv (-\mathsf{ab})k^{5+p_{12}}\int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\,(-\tau_1)^{p_1}(-\tau_2)^{p_2}e^{\mathrm{i}\mathsf{a}k\tau_1+\mathrm{i}\mathsf{b}k\tau_2}\frac{D^{(\rm s)}_{\mathsf{ab}}(k;\tau_1,\tau_2)}{H^2}, \tag{76}$$

This is the seed $\widetilde{\mathcal{I}}^{p_1p_2}_{\mathsf{ab}}(1,1)$ of Ref. [3], with H restored explicitly. All seeds below have mass index $\tilde{\nu}$. The closed expressions are

$$\mathcal{M}^{p_1p_2}_{\pm\mp} = \frac{e^{\mp\mathrm{i}\pi\bar{p}_{12}/2}}{2^{5+p_{12}}}\prod_{j=1}^{2}\frac{\Gamma(5/2+p_j-\mathrm{i}\tilde{\nu})\Gamma(5/2+p_j+\mathrm{i}\tilde{\nu})}{\Gamma(3+p_j)}, \tag{77}$$

$$\begin{aligned}\mathcal{M}^{p_1p_2}_{\pm\pm} &= \frac{\pm\mathrm{i}e^{\mp\mathrm{i}\pi p_{12}/2}e^{-\pi\tilde{\nu}}}{2^{5+p_{12}}}\prod_{j=1}^{2}\frac{\Gamma(5/2+p_j-\mathrm{i}\tilde{\nu})\Gamma(5/2+p_j+\mathrm{i}\tilde{\nu})}{\Gamma(3+p_j)} \\ &\quad - \frac{e^{\mp\mathrm{i}\pi p_{12}/2}}{2^{5+p_{12}}}\Gamma(5+p_{12})\Gamma(5/2+p_1\pm\mathrm{i}\tilde{\nu})\Gamma(5/2+p_2\pm\mathrm{i}\tilde{\nu}) \\ &\quad \times {}_3\widetilde{F}_2\left[\begin{matrix}5+p_{12},\ 1/2\pm\mathrm{i}\tilde{\nu},\ 1\\ 7/2+p_1\pm\mathrm{i}\tilde{\nu},\ 7/2+p_2\pm\mathrm{i}\tilde{\nu}\end{matrix}\middle|1\right]. \end{aligned} \tag{78}$$

Here ${}_3\widetilde{F}_2(a,b,c;d,e;1) = {}_3F_2(a,b,c;d,e;1)/[\Gamma(d)\Gamma(e)]$ is the regularized hypergeometric function.

Set $p_\epsilon = -5/2+\epsilon+\mathrm{i}\eta\tilde{\nu}$. The two transverse coefficients are

$$\mathcal{P}^{(2),\pm}_{\eta}(\tilde{\nu}) = 2^{2\mathrm{i}\eta\tilde{\nu}}\underset{\epsilon=0}{\rm FP}\sum_{\mathsf{a},\mathsf{b}=\pm1}\left[\mathcal{M}^{p_\epsilon,p_\epsilon}_{\mathsf{ab}} + \mathrm{i}\mathsf{a}\mathcal{M}^{p_\epsilon+1,p_\epsilon}_{\mathsf{ab}} + \mathrm{i}\mathsf{b}\mathcal{M}^{p_\epsilon,p_\epsilon+1}_{\mathsf{ab}} - \mathsf{ab}\mathcal{M}^{p_\epsilon+1,p_\epsilon+1}_{\mathsf{ab}}\right]. \tag{79}$$

The longitudinal coefficient can be written using the same transverse combination:

$$\begin{aligned}\mathcal{P}^{(2),L}_{\eta}(\tilde{\nu}) &= \frac{\mathcal{P}^{(1)}_{\eta}(\tilde{\nu}) + (1/2+\mathrm{i}\eta\tilde{\nu})^2\mathcal{P}^{(2),+}_{\eta}(\tilde{\nu})}{\tilde{\nu}^2+1/4} \\ &\quad + \frac{2^{2\mathrm{i}\eta\tilde{\nu}}}{\tilde{\nu}^2+1/4}\underset{\epsilon=0}{\rm FP}\sum_{\mathsf{a},\mathsf{b}=\pm1}\Big\{(1/2+\mathrm{i}\eta\tilde{\nu})\left[\mathcal{M}^{p_\epsilon+2,p_\epsilon}_{\mathsf{ab}} + \mathrm{i}\mathsf{b}\mathcal{M}^{p_\epsilon+2,p_\epsilon+1}_{\mathsf{ab}}\right. \\ &\qquad \left. + \mathcal{M}^{p_\epsilon,p_\epsilon+2}_{\mathsf{ab}} + \mathrm{i}\mathsf{a}\mathcal{M}^{p_\epsilon+1,p_\epsilon+2}_{\mathsf{ab}}\right] + \mathcal{M}^{p_\epsilon+2,p_\epsilon+2}_{\mathsf{ab}}\Big\}. \end{aligned} \tag{80}$$

The operator $\underset{\epsilon=0}{\rm FP}$ selects the constant coefficient of the Laurent expansion after assembling the indicated seed sum. The first term proportional to $\mathcal{P}^{(1)}_{\eta}$ in (80) is the contact contribution from differentiating the time-ordering step functions.

A.4 Box Coefficients

A.4.1 Polarization sums

Let $\lambda_1, \lambda_3 = L, \pm$ denote the polarizations of the hard lines with momenta $\mathbf{k}_1$ and $\mathbf{k}_3$. With $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, define the soft-polarization sum by

$$\sum_{\sigma_1,\sigma_2=L,\pm} [\beta(x)]^{\delta_{\sigma_1 L}+\delta_{\sigma_2 L}} \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \left(\frac{q}{2}\right)^{-2x} \left(\frac{K}{2}\right)^{-2x} (e^{(\lambda_1)*}_{\mathbf{k}_1} \cdot e^{(\sigma_1)}_{\mathbf{K}})(e^{(\sigma_1)*}_{\mathbf{K}} \cdot e^{(\lambda_3)}_{\mathbf{k}_3})$$
$$\times (e^{(\lambda_3)*}_{\mathbf{k}_3} \cdot e^{(\sigma_2)}_{\mathbf{q}})(e^{(\sigma_2)*}_{\mathbf{q}} \cdot e^{(\lambda_1)}_{\mathbf{k}_1}) = \frac{4^{2x} k_s^{3-4x}}{(4\pi)^{3/2}} \mathcal{I}^{\lambda_1\lambda_3}_{\mathrm{Box}}(x, \theta_1, \theta_3, \phi_3). \tag{81}$$

The angles are defined by the convention stated below (1). The two transverse hard-line coefficients $\mathcal{P}^{(2),+}_\eta = \mathcal{P}^{(2),-}_\eta$ are equal, so it suffices to retain three groups:

$$\mathcal{I}^{LH}_{\mathrm{Box}} \equiv \sum_{\lambda=\pm} \left(\mathcal{I}^{L\lambda}_{\mathrm{Box}} + \mathcal{I}^{\lambda L}_{\mathrm{Box}}\right), \qquad \mathcal{I}^{HH}_{\mathrm{Box}} \equiv \sum_{\lambda_1,\lambda_3=\pm} \mathcal{I}^{\lambda_1\lambda_3}_{\mathrm{Box}}, \tag{82}$$

together with $\mathcal{I}^{LL}_{\mathrm{Box}}$. Here H in a polarization superscript denotes the sum over transverse helicities.

For $x = -\mathrm{i}\tilde{\nu}$, the longitudinal result is

$$\begin{aligned}
&\mathcal{I}^{LL}_{\mathrm{Box}}(-\mathrm{i}\tilde{\nu}, \theta_1, \theta_3, \phi_3) \\
&= -\frac{4^{-3-2\mathrm{i}\tilde{\nu}}(2\tilde{\nu} - 3\mathrm{i})\Gamma(-3/2 - 2\mathrm{i}\tilde{\nu})\Gamma(1 + 2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu} - \mathrm{i})(\tilde{\nu} - 2\mathrm{i})(2\tilde{\nu} - \mathrm{i})} \\
&\quad \times \Big\{ -16 - 19\mathrm{i}\tilde{\nu} + 6\tilde{\nu}^2 + \tilde{\nu}(2\tilde{\nu} - \mathrm{i})\Big[\cos(2\theta_1) + \cos(2\theta_3) + 3\cos(2\theta_1)\cos(2\theta_3)\Big] \\
&\qquad + 4(2\tilde{\nu}^2 - 5\mathrm{i}\tilde{\nu} - 3)\cos(2\phi_3)\sin^2\theta_1 \sin^2\theta_3 \\
&\qquad - 8\tilde{\nu}(\tilde{\nu} - \mathrm{i})\cos\phi_3 \sin(2\theta_1)\sin(2\theta_3)\Big\}.
\end{aligned} \tag{83}$$

Reducing the mixed and transverse results to the same Gamma-function basis gives

$$\begin{aligned}
&\mathcal{I}^{LH}_{\mathrm{Box}}(-\mathrm{i}\tilde{\nu}, \theta_1, \theta_3, \phi_3) \\
&= -\frac{2 \cdot 4^{-3-2\mathrm{i}\tilde{\nu}}(2\tilde{\nu} - 3\mathrm{i})\Gamma(-3/2 - 2\mathrm{i}\tilde{\nu})\Gamma(1 + 2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu} - \mathrm{i})(\tilde{\nu} - 2\mathrm{i})(2\tilde{\nu} - \mathrm{i})} \\
&\quad \times \Big\{ -32 - 37\mathrm{i}\tilde{\nu} + 10\tilde{\nu}^2 - \tilde{\nu}(2\tilde{\nu} - \mathrm{i})\Big[\cos(2\theta_1) + \cos(2\theta_3) + 3\cos(2\theta_1)\cos(2\theta_3)\Big] \\
&\qquad - 4(2\tilde{\nu}^2 - 5\mathrm{i}\tilde{\nu} - 3)\cos(2\phi_3)\sin^2\theta_1 \sin^2\theta_3 \\
&\qquad + 8\tilde{\nu}(\tilde{\nu} - \mathrm{i})\cos\phi_3 \sin(2\theta_1)\sin(2\theta_3)\Big\},
\end{aligned} \tag{84}$$

$$\begin{aligned}
&\mathcal{I}^{HH}_{\mathrm{Box}}(-\mathrm{i}\tilde{\nu}, \theta_1, \theta_3, \phi_3) \\
&= -\frac{4^{-3-2\mathrm{i}\tilde{\nu}}(2\tilde{\nu} - 3\mathrm{i})\Gamma(-3/2 - 2\mathrm{i}\tilde{\nu})\Gamma(1 + 2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu} - \mathrm{i})(\tilde{\nu} - 2\mathrm{i})(2\tilde{\nu} - \mathrm{i})} \\
&\quad \times \Big\{ -64 - 75\mathrm{i}\tilde{\nu} + 22\tilde{\nu}^2 + \tilde{\nu}(2\tilde{\nu} - \mathrm{i})\Big[\cos(2\theta_1) + \cos(2\theta_3) + 3\cos(2\theta_1)\cos(2\theta_3)\Big] \\
&\qquad + 4(2\tilde{\nu}^2 - 5\mathrm{i}\tilde{\nu} - 3)\cos(2\phi_3)\sin^2\theta_1 \sin^2\theta_3 \\
&\qquad - 8\tilde{\nu}(\tilde{\nu} - \mathrm{i})\cos\phi_3 \sin(2\theta_1)\sin(2\theta_3)\Big\}.
\end{aligned} \tag{85}$$

These expressions satisfy the relations implied by polarization completeness and (74):

$$\mathcal{I}^{LH}_{\rm Box} = \frac{2}{3}\mathcal{I}_{\rm Bub} - 2\mathcal{I}^{LL}_{\rm Box}, \tag{86}$$

$$\mathcal{I}^{HH}_{\rm Box} = \frac{1}{3}\mathcal{I}_{\rm Bub} + \mathcal{I}^{LL}_{\rm Box}. \tag{87}$$

Indeed, all angular terms cancel in $\mathcal{I}^{LH}_{\rm Box} + 2\mathcal{I}^{LL}_{\rm Box}$ and $\mathcal{I}^{HH}_{\rm Box} - \mathcal{I}^{LL}_{\rm Box}$. The three groups therefore also satisfy $\mathcal{I}^{LL}_{\rm Box} + \mathcal{I}^{LH}_{\rm Box} + \mathcal{I}^{HH}_{\rm Box} = \mathcal{I}_{\rm Bub}$.

The dimensionless box coefficient is consequently

$$\begin{aligned} C^{\rm Box}(\tilde{\nu},\theta_1,\theta_3,\phi_3) &= \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}}\Big\{[\mathcal{P}^{(2),L}_+]^2\mathcal{I}^{LL}_{\rm Box} + \mathcal{P}^{(2),L}_+\mathcal{P}^{(2),+}_+\mathcal{I}^{LH}_{\rm Box} + [\mathcal{P}^{(2),+}_+]^2\mathcal{I}^{HH}_{\rm Box}\Big\} \\ &= \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}}\Big\{[\mathcal{P}^{(2),L}_+ - \mathcal{P}^{(2),+}_+]^2\mathcal{I}^{LL}_{\rm Box} + \frac{1}{3}\mathcal{P}^{(2),+}_+[2\mathcal{P}^{(2),L}_+ + \mathcal{P}^{(2),+}_+]\mathcal{I}_{\rm Bub}\Big\}, \end{aligned} \tag{88}$$

The opposite nonlocal branch follows by complex conjugation.

A.4.2 Angular projections

The five functions in (37) can be read off from (88), or equivalently obtained from the following projections:

$$A_{1,g}(\tilde{\nu}) = \frac{1}{2}\left[C^{\rm Box}(\tilde{\nu},\tfrac{\pi}{2},\tfrac{\pi}{2},0) + C^{\rm Box}(\tilde{\nu},\tfrac{\pi}{2},\tfrac{\pi}{2},\tfrac{\pi}{2})\right], \tag{89}$$

$$A_{2,g}(\tilde{\nu}) = C^{\rm Box}(\tilde{\nu},\tfrac{\pi}{2},0,0) - A_{1,g}(\tilde{\nu}), \tag{90}$$

$$A_{3,g}(\tilde{\nu}) = C^{\rm Box}(\tilde{\nu},0,0,0) - A_{1,g}(\tilde{\nu}) - 2A_{2,g}(\tilde{\nu}), \tag{91}$$

$$B_g(\tilde{\nu}) = \frac{1}{2}\left[C^{\rm Box}(\tilde{\nu},\tfrac{\pi}{4},\tfrac{\pi}{4},0) - C^{\rm Box}(\tilde{\nu},\tfrac{\pi}{4},\tfrac{\pi}{4},\pi)\right], \tag{92}$$

$$D_g(\tilde{\nu}) = \frac{1}{2}\left[C^{\rm Box}(\tilde{\nu},\tfrac{\pi}{2},\tfrac{\pi}{2},0) - C^{\rm Box}(\tilde{\nu},\tfrac{\pi}{2},\tfrac{\pi}{2},\tfrac{\pi}{2})\right]. \tag{93}$$

B Top Quark Integrals and Coefficients

B.1 Propagators and Spinor Contractions

We use $\sigma^a = (\mathbf{1}_2, \boldsymbol{\sigma})$, $\bar{\sigma}^a = (\mathbf{1}_2, -\boldsymbol{\sigma})$, and $\psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta$ with $\epsilon_{12} = 1$, $\epsilon^{12} = -1$. The helicity projectors are $P_s = h^s h^{s\dagger} = (\mathbf{1}_2 + s\sigma_k)/2$. For the rescaled field, the mode expansion is

$$\psi(\tau,\mathbf{x}) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\sum_{s=\pm1}\left[h^s u_s b^s_{\mathbf{k}} e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}} + \boldsymbol{\epsilon} h^{s*} v^*_s b^{s\dagger}_{\mathbf{k}} e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\right], \tag{94}$$

where $\{b^s_{\mathbf{k}}, b^{s'\dagger}_{\mathbf{k}'}\} = (2\pi)^3\delta^{ss'}\delta^3(\mathbf{k}-\mathbf{k}')$. The modes in (41) satisfy $\mathrm{i}\partial_\tau u_s + sku_s = am_t v_s$ and $\mathrm{i}\partial_\tau v_s - skv_s = am_t u_s$. With $u_{sj} = u_s(k,\tau_j)$ and $v_{sj} = v_s(k,\tau_j)$, the three unordered contractions are

$$D_{-+} = \sum_s u_{s1}u^*_{s2}P_s, \qquad \widehat{D}_{-+} = -\sum_s u_{s1}v^*_{s2}P_s\boldsymbol{\epsilon}, \qquad \check{D}_{-+} = \sum_s v_{s1}u^*_{s2}\boldsymbol{\epsilon}P_s. \tag{95}$$

The signs follow from $h^s(\boldsymbol{\epsilon}h^{s*})^T = -P_s\boldsymbol{\epsilon}$.

For the seed reduction it is useful to express these contractions through the scalar-valued Whittaker kernel [3]. With $X = -k\tau_1$, $Y = -k\tau_2$,

$$D^{\mathrm{sc},\kappa}_{-+}(k;\tau_1,\tau_2) = \frac{e^{\mathrm{i}\pi\kappa}}{2k} W_{\kappa,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}X) W_{-\kappa,\mathrm{i}\tilde{\nu}}(2\mathrm{i}Y),$$
$$D^{\mathrm{sc},\kappa}_{+-}(k;\tau_1,\tau_2) = \frac{e^{\mathrm{i}\pi\kappa}}{2k} W_{-\kappa,\mathrm{i}\tilde{\nu}}(2\mathrm{i}X) W_{\kappa,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}Y). \tag{96}$$

The other two SK components follow from (43). In the notation of the reference, $\kappa = \mathrm{i}h\tilde{\mu}$; both Wightman kernels are analytically continued to $\kappa = \pm 1/2$. Complex conjugation of the continued parameter is not performed separately. Write $D^{\mathrm{sc},\pm} = D^{\mathrm{sc},\pm 1/2}$. The normalization $1/(2k)$ differs from that of the covariant scalar propagator in (6).

Whittaker recurrences then give, for all four SK choices,

$$D_{\mathsf{ab}} = \frac{\mathrm{i}k}{\sqrt{XY}} \left[(-X\partial_X - \mathrm{i}X + \tfrac{1}{2}) D^{\mathrm{sc},-}_{\mathsf{ab}} P_- - (X\partial_X - \mathrm{i}X - \tfrac{1}{2}) D^{\mathrm{sc},+}_{\mathsf{ab}} P_+ \right],$$
$$\widehat{D}_{\mathsf{ab}} = -\frac{\tilde{\nu}k}{\sqrt{XY}} [D^{\mathrm{sc},-}_{\mathsf{ab}} P_+ + D^{\mathrm{sc},+}_{\mathsf{ab}} P_-] \boldsymbol{\epsilon},$$
$$\check{D}_{\mathsf{ab}} = \frac{\tilde{\nu}k}{\sqrt{XY}} \boldsymbol{\epsilon} [D^{\mathrm{sc},+}_{\mathsf{ab}} P_+ + D^{\mathrm{sc},-}_{\mathsf{ab}} P_-]. \tag{97}$$

The derivatives act only on the kernels. Their Wightman values coincide at equal times, so these first derivatives generate no delta function from the SK step functions. The remaining field-type block is fixed by fermionic reversal:

$$D^{\mathrm{L}\bar{\mathrm{L}}} = D, \quad D^{\mathrm{LL}} = \widehat{D}, \quad D^{\bar{\mathrm{L}}\bar{\mathrm{L}}} = \check{D}, \quad D^{\bar{\mathrm{L}}\mathrm{L}}_{\mathsf{ab}}(\mathbf{k};\tau_1,\tau_2) = -D^{T}_{\mathsf{ba}}(-\mathbf{k};\tau_2,\tau_1). \tag{98}$$

Both the reversed momentum and the minus sign are required. Expanding

$$W_{\kappa,\mathrm{i}\tilde{\nu}}(z) = \sum_{\eta=\pm 1} \frac{\Gamma(-2\mathrm{i}\eta\tilde{\nu})}{\Gamma(\frac{1}{2} - \mathrm{i}\eta\tilde{\nu} - \kappa)} z^{1/2+\mathrm{i}\eta\tilde{\nu}} \left[1 - \frac{\kappa z}{1 + 2\mathrm{i}\eta\tilde{\nu}} + O(z^2) \right] \tag{99}$$

in (97) gives (45).

Two scalar hard coefficients suffice for the vertex-type sums. At hard momentum p, set $X = -p\tau_1$, $Y = -p\tau_2$ and define

$$F_{0;\mathsf{ab}}(p;\tau_1,\tau_2) = \frac{\mathrm{i}p}{2\sqrt{XY}} \left[(X\partial_X - \mathrm{i}X - \tfrac{1}{2} - \mathrm{i}\tilde{\nu}) D^{\mathrm{sc},+}_{\mathsf{ab}} + (-X\partial_X - \mathrm{i}X + \tfrac{1}{2} + \mathrm{i}\tilde{\nu}) D^{\mathrm{sc},-}_{\mathsf{ab}} \right],$$
$$F_{1;\mathsf{ab}}(p;\tau_1,\tau_2) = \frac{\mathrm{i}p}{2\sqrt{XY}} \left[(-X\partial_X - \mathrm{i}X + \tfrac{1}{2} - \mathrm{i}\tilde{\nu}) D^{\mathrm{sc},-}_{\mathsf{ab}} - (X\partial_X - \mathrm{i}X - \tfrac{1}{2} + \mathrm{i}\tilde{\nu}) D^{\mathrm{sc},+}_{\mathsf{ab}} \right]. \tag{100}$$

Every kernel has arguments $(p;\tau_1,\tau_2)$. In particular $F_{0;\mathsf{ab}}(p;\tau_1,\tau_2) = F_{0;\mathsf{ba}}(p;\tau_2,\tau_1)$. These dimensionless functions describe the hard spin matrices; the calligraphic $\mathcal{K}$ below is reserved for the time seed.

For fixed external labels, the four bilinear insertions have 48 connected Wick pairings: 16 for each of the three undirected cycles. The four factors of $1/2$ in the interaction cancel these 16 pairings. Expressing each undirected cycle as its two directed orientations therefore assigns $-g_t^4/2$ to each orientation in one Majorana sector. The closed-loop sign and the restoration of $2N_c$ give (46).

B.2 Soft Expansion and Loop Integrals

For the $\eta = +1$ branch, factor the two soft endpoints as

$$D^{rs}_{\mathrm{NonL},+}(p;\tau,\tau') \simeq -\frac{c_+}{2}(p^2\tau\tau')^{\mathrm{i}\tilde{\nu}} L_r(p,\tau)\sigma_p U_s(p,\tau'), \tag{101}$$

where, through first order,

$$\begin{aligned} L_{\mathrm{L}} &= \mathbf{1}_2 + \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{p}}{1+2\mathrm{i}\tilde{\nu}}, & L_{\overline{\mathrm{L}}} &= \boldsymbol{\epsilon}\left(\mathbf{1}_2 - \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{p}}{1+2\mathrm{i}\tilde{\nu}}\right), \\ U_{\mathrm{L}} &= \left(\mathbf{1}_2 + \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{p}}{1+2\mathrm{i}\tilde{\nu}}\right)\boldsymbol{\epsilon}, & U_{\overline{\mathrm{L}}} &= \mathbf{1}_2 - \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{p}}{1+2\mathrm{i}\tilde{\nu}}. \end{aligned} \tag{102}$$

The matrices in (48) are the vertex-type sums

$$\begin{aligned} V_{12} &= -\frac{1}{2}\sum_{r_1,r_2} U_{r_1}(q,\tau_1)J_{r_1}D^{r_1r_2}_{\mathsf{a}_1\mathsf{a}_2}(\mathbf{k}_L+\boldsymbol{\ell};\tau_1,\tau_2)J_{r_2}L_{r_2}(K,\tau_2), \\ V_{21} &= -\frac{1}{2}\sum_{r_1,r_2} U_{r_2}(q,\tau_2)J_{r_2}D^{r_2r_1}_{\mathsf{a}_2\mathsf{a}_1}(-\mathbf{k}_L+\boldsymbol{\ell};\tau_2,\tau_1)J_{r_1}L_{r_1}(K,\tau_1). \end{aligned} \tag{103}$$

The right matrices follow by $(L,1,2,q,K)\to(R,3,4,K,q)$. The factors $-1/2$ in these definitions reproduce the normalization of the two cut lines. Summing the four vertex choices at one hard end first gives $V_{12}^{(0)} = -F_0\boldsymbol{\sigma}\cdot\mathbf{n}_L$. Reversal changes the direction but leaves F_0 unchanged, proving (50).

At first order, the hard-line correction in V_{12} is

$$-\frac{1}{k_L}\boldsymbol{\sigma}\cdot\left[F_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\ell} + (p\partial_p - 1)F_{0;\mathsf{a}_1\mathsf{a}_2}(p;\tau_1,\tau_2)\big|_{p=k_L}\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell})\right], \tag{104}$$

while the two soft endpoints supply

$$\frac{\mathrm{i}}{1+2\mathrm{i}\tilde{\nu}}\boldsymbol{\sigma}\cdot\left[\tau_1 F_{1;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\mathbf{q} + \tau_2 F_{1;\mathsf{a}_2\mathsf{a}_1}(k_L;\tau_2,\tau_1)\mathbf{K}\right]. \tag{105}$$

The reversed ordering has the same hard correction and interchanges the coefficients of $\mathbf{q}$ and $\mathbf{K}$. Since $\mathbf{q}+\mathbf{K} = 2\boldsymbol{\ell}$, adding the two orderings gives (51), with

$$\begin{aligned} \Pi_{1;\mathsf{ab}}(k;\tau_1,\tau_2) &= F_{0;\mathsf{ab}}(k;\tau_1,\tau_2) - \frac{\mathrm{i}k}{1+2\mathrm{i}\tilde{\nu}}[\tau_1 F_{1;\mathsf{ab}}(k;\tau_1,\tau_2) + \tau_2 F_{1;\mathsf{ba}}(k;\tau_2,\tau_1)], \\ \Pi_{2;\mathsf{ab}}(k;\tau_1,\tau_2) &= (p\partial_p - 1)F_{0;\mathsf{ab}}(p;\tau_1,\tau_2)\big|_{p=k}. \end{aligned} \tag{106}$$

The momentum derivative is at fixed times. No equality between the two F_1 terms is assumed. This addition involves two physical diagrams and does not average their time or SK labels. Using $\mathrm{tr}_2[(\boldsymbol{\sigma}\cdot\mathbf{a})(\boldsymbol{\sigma}\cdot\mathbf{b})(\boldsymbol{\sigma}\cdot\mathbf{c})(\boldsymbol{\sigma}\cdot\mathbf{d})] = 2[(\mathbf{a}\cdot\mathbf{b})(\mathbf{c}\cdot\mathbf{d}) - (\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d}) + (\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})]$, the sum of the four first-order traces becomes

$$\begin{aligned} &\mathrm{tr}_2[(V_{12}^{(1)} + V_{21}^{(1)})\sigma_K(V_{34}^{(1)} + V_{43}^{(1)})\sigma_q] \\ &\quad = \frac{8}{k_L k_R q K}\left[2(\mathbf{v}_L\cdot\boldsymbol{\ell})(\mathbf{v}_R\cdot\boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L\cdot\mathbf{k}_s)(\mathbf{v}_R\cdot\mathbf{k}_s) - (\mathbf{v}_L\cdot\mathbf{v}_R)(\ell^2 - k_s^2/4)\right]. \end{aligned} \tag{107}$$

To reduce the polynomial numerator in (52), measure all loop momenta in units of k_s and set $u = q^2/k_s^2$, $v = K^2/k_s^2$, and $\rho = 1/2 - \mathrm{i}\tilde{\nu}$. Then

$$\frac{\ell^2}{k_s^2} = \frac{u+v}{2} - \frac{1}{4}, \qquad \frac{\ell_\parallel}{k_s} = \frac{v-u}{2}, \qquad \ell_\perp^2 = \ell^2 - \ell_\parallel^2. \tag{108}$$

Azimuthal averaging replaces two transverse components by $\ell_\perp^2 \delta_{ij}^\perp/2$ and four by $\ell_\perp^4(\delta_{ij}^\perp\delta_{mn}^\perp + \delta_{im}^\perp\delta_{jn}^\perp + \delta_{in}^\perp\delta_{jm}^\perp)/8$. Each resulting monomial is integrated using

$$u^m v^n \;\longmapsto\; \frac{f_0(\rho-m,\rho-n)}{(4\pi)^{3/2}}, \qquad \frac{f_0(\rho-m,\rho-n)}{f_0(\rho,\rho)} = \frac{(3/2-\rho)_m(3/2-\rho)_n(1-\rho)_m(1-\rho)_n}{(5/2-2\rho)_{m+n}(3-2\rho)_{m+n}}. \tag{109}$$

Here $(a)_m$ is the Pochhammer symbol and $m+n \le 4$. The coefficients defined in (52) are

$$\begin{aligned}
\mathcal{I}^{11}_{t,\mathrm{Box}} &= 3\mathcal{I}^{12}_{t,\mathrm{Box}} = 3\mathcal{I}^{21}_{t,\mathrm{Box}},\\
\mathcal{I}^{12}_{t,\mathrm{Box}} &= \frac{f_0(\rho,\rho)}{(4\pi)^{3/2}}\frac{(\rho-1)(2\rho-3)}{8(\rho-2)(4\rho-7)(4\rho-5)},\\
\mathcal{I}^{22}_{t,\mathrm{Box}} &= \frac{f_0(\rho,\rho)}{32(4\pi)^{3/2}(\rho-3)(\rho-2)(4\rho-7)(4\rho-5)}\Big\{(\rho-1)(2\rho-3)(2\rho-5)\\
&\quad - (\rho-1)(2\rho-3)^2[\cos^2\theta_1 + \cos^2\theta_3 - 3\cos^2\theta_1\cos^2\theta_3]\\
&\quad - 2(\rho-1)(\rho-2)(2\rho-3)\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3\\
&\quad + (\rho-2)(2\rho-5)(2\rho-3)\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3)\Big\}.
\end{aligned} \tag{110}$$

The scalar convolution is first defined in its convergence domain and continued after the shifted moments are combined. Extending the soft region beyond $q \sim k_s$ produces ultraviolet power divergences; analytic continuation isolates the nonanalytic coefficient, while polynomial terms belong to matching with the hard region. This prescription does not determine contact-term renormalization.

B.3 Whittaker Seed and Time Integrals

We denote the Whittaker seed $\mathcal{I}$ of Ref. [3] by $\mathcal{K}$. For equal external energies k, internal magnitude $p = rk$, and $u = 2r/(1+r)$, its diagonal restriction is defined by

$$\mathcal{K}^{\kappa;P,Q}_{\mathsf{ab}}(u,u) \equiv (-\mathsf{ab})p^{3+P+Q}\int_{-\infty}^{0} \mathrm{d}\tau_1\mathrm{d}\tau_2\,(-\tau_1)^P(-\tau_2)^Q e^{\mathrm{i}\mathsf{a}k\tau_1+\mathrm{i}\mathsf{b}k\tau_2} D^{\mathrm{sc},\kappa}_{\mathsf{ab}}(p;\tau_1,\tau_2). \tag{111}$$

The phase $e^{\mathrm{i}\pi\kappa}$ is already included in the scalar kernel. In terms of $x = -k\tau_1$, $y = -k\tau_2$, its time moment is

$$\int_0^\infty \mathrm{d}x\mathrm{d}y\, x^P y^Q e^{-\mathrm{i}\mathsf{a}x-\mathrm{i}\mathsf{b}y} k D^{\mathrm{sc},\kappa}_{\mathsf{ab}}(rk;-x/k,-y/k) = -\mathsf{ab}\, r^{-3-P-Q}\mathcal{K}^{\kappa;P,Q}_{\mathsf{ab}}(u,u). \tag{112}$$

There is no shift of the mass index $\tilde{\nu}$ in this reduction.

For brevity, the seed formulas below use $\Gamma[z_1,\dots,z_n]=\prod_i\Gamma(z_i)$. At $r=1$, write $\mathcal{K}^{\kappa;P,Q}_{\mathsf{ab}}\equiv\mathcal{K}^{\kappa;P,Q}_{\mathsf{ab}}(1,1)$. The closed expressions are

$$\mathcal{K}^{\kappa;P,Q}_{\mathsf{a},-\mathsf{a}}=\frac{e^{-\mathsf{i}\mathsf{a}\pi(P-Q)/2+\mathrm{i}\pi\kappa}}{2^{3+P+Q}}\frac{\Gamma[\frac{3}{2}+P-\mathsf{i}\tilde{\nu},\frac{3}{2}+P+\mathsf{i}\tilde{\nu},\frac{3}{2}+Q-\mathsf{i}\tilde{\nu},\frac{3}{2}+Q+\mathsf{i}\tilde{\nu}]}{\Gamma[2+P+\mathsf{a}\kappa,2+Q-\mathsf{a}\kappa]}, \tag{113}$$

$$\begin{aligned}\mathcal{K}^{\kappa;P,Q}_{\mathsf{aa}}=&\frac{-\mathsf{a}\mathsf{i}e^{-\mathsf{i}\mathsf{a}\pi(P+Q)/2}}{2^{3+P+Q}}\Gamma[\tfrac{3}{2}+P+\mathsf{a}\mathsf{i}\tilde{\nu},\tfrac{3}{2}+Q+\mathsf{a}\mathsf{i}\tilde{\nu}]\\&\times\Bigg\{\frac{e^{2\mathrm{i}\pi\kappa}+e^{-2\pi\tilde{\nu}}}{2\pi}\frac{\Gamma[\frac{3}{2}+P-\mathsf{a}\mathsf{i}\tilde{\nu},\frac{3}{2}+Q-\mathsf{a}\mathsf{i}\tilde{\nu},\frac{1}{2}+\mathsf{a}\kappa-\mathsf{i}\tilde{\nu},\frac{1}{2}+\mathsf{a}\kappa+\mathsf{i}\tilde{\nu}]}{\Gamma[2+P+\mathsf{a}\kappa,2+Q+\mathsf{a}\kappa]}\\&+\mathsf{a}\mathsf{i}\Gamma(3+P+Q)\,{}_3\widetilde{F}_2\left[\begin{matrix}3+P+Q,\ \frac{1}{2}-\mathsf{a}\kappa+\mathsf{a}\mathsf{i}\tilde{\nu},\ 1\\ \frac{5}{2}+P+\mathsf{a}\mathsf{i}\tilde{\nu},\ \frac{5}{2}+Q+\mathsf{a}\mathsf{i}\tilde{\nu}\end{matrix}\middle|1\right]\Bigg\}.\end{aligned} \tag{114}$$

The regularized hypergeometric function has the same definition as in App. A.3, namely

$${}_3\widetilde{F}_2(a,b,c;d,e;1)=\frac{{}_3F_2(a,b,c;d,e;1)}{\Gamma(d)\Gamma(e)}.$$

At the required powers these formulas denote analytic continuation of the complete seed combination.

Only two scalar time moments are needed to express all three hard coefficients. Define

$$\begin{aligned}\mathcal{H}_j(\alpha,\beta;r)=&\sum_{\mathsf{a},\mathsf{b}=\pm1}\mathsf{a}\mathsf{b}\int_0^\infty\mathrm{d}x\mathrm{d}y\,x^{\alpha-1}y^{\beta-1}(1+\mathsf{i}\mathsf{a}x)(1+\mathsf{i}\mathsf{b}y)e^{-\mathsf{i}\mathsf{a}x-\mathsf{i}\mathsf{b}y}\\&\times F_{j;\mathsf{ab}}(rk;-x/k,-y/k),\qquad j=0,1.\end{aligned} \tag{115}$$

Scale invariance makes the right-hand side independent of k. Expanding the external polynomials with indices $m,n=0,1$, put $P=\alpha-3/2+m$, $Q=\beta-3/2+n$. Integration by parts sends $x\partial_x$ to $-(P+1)+\mathsf{i}\mathsf{a}x$. Substituting (112) therefore gives directly

$$\begin{aligned}\mathcal{H}_0(\alpha,\beta;r)=&-\frac{\mathsf{i}}{2}\sum_{\mathsf{a},\mathsf{b}=\pm1}\sum_{m,n=0}^{1}(\mathsf{i}\mathsf{a})^m(\mathsf{i}\mathsf{b})^n r^{-3-P-Q}\Big\{(P+\tfrac{3}{2}+\mathsf{i}\tilde{\nu})[\mathcal{K}^{-;P,Q}_{\mathsf{ab}}-\mathcal{K}^{+;P,Q}_{\mathsf{ab}}]\\&+\frac{\mathsf{i}(\mathsf{a}-r)}{r}\mathcal{K}^{+;P+1,Q}_{\mathsf{ab}}-\frac{\mathsf{i}(\mathsf{a}+r)}{r}\mathcal{K}^{-;P+1,Q}_{\mathsf{ab}}\Big\},\end{aligned} \tag{116}$$

$$\begin{aligned}\mathcal{H}_1(\alpha,\beta;r)=&-\frac{\mathsf{i}}{2}\sum_{\mathsf{a},\mathsf{b}=\pm1}\sum_{m,n=0}^{1}(\mathsf{i}\mathsf{a})^m(\mathsf{i}\mathsf{b})^n r^{-3-P-Q}\Big\{(P+\tfrac{3}{2}-\mathsf{i}\tilde{\nu})[\mathcal{K}^{-;P,Q}_{\mathsf{ab}}+\mathcal{K}^{+;P,Q}_{\mathsf{ab}}]\\&-\frac{\mathsf{i}(\mathsf{a}-r)}{r}\mathcal{K}^{+;P+1,Q}_{\mathsf{ab}}-\frac{\mathsf{i}(\mathsf{a}+r)}{r}\mathcal{K}^{-;P+1,Q}_{\mathsf{ab}}\Big\}.\end{aligned} \tag{117}$$

Every seed here has arguments (u,u), and its superscript $\pm$ means $\kappa=\pm1/2$, independently of the SK labels. The integration by parts is performed first where the regulated boundary terms vanish, with the Bunch–Davies early-time prescription.

The bootstrap equation supplies the hard-momentum derivative at the folded point without

introducing another integral:

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}r}\mathcal{K}^{\kappa;P,Q}_{\mathsf{ab}}(u(r),u(r))\Big|_{r=1}\\
&\quad=\frac{1}{2}\left[\frac{\tilde\nu^2+(P+\frac{3}{2})^2}{2+P+\mathsf{a}\kappa}+\frac{\tilde\nu^2+(Q+\frac{3}{2})^2}{2+Q+\mathsf{b}\kappa}\right]\mathcal{K}^{\kappa;P,Q}_{\mathsf{ab}}(1,1)\\
&\qquad+\frac{\delta_{\mathsf{ab}}}{2}\left[\frac{1}{2+P+\mathsf{a}\kappa}+\frac{1}{2+Q+\mathsf{b}\kappa}\right]e^{-\mathrm{i}\mathsf{a}\pi(P+Q)/2}2^{-3-P-Q}\Gamma(3+P+Q).
\end{aligned}\tag{118}$$

Both the explicit powers of r and the factors $(\mathsf{a}\pm r)/r$ in (116) must also be differentiated. The coefficients used in the main text are

$$\begin{aligned}
\mathcal{P}_{t,1}&=\lim_{\delta\to0}\left[\mathcal{H}_0(\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;1)+\frac{2\mathrm{i}}{1+2\mathrm{i}\tilde\nu}\mathcal{H}_1(1+\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;1)\right],\\
\mathcal{P}_{t,2}&=\lim_{\delta\to0}\,(\partial_r-1)\mathcal{H}_0(\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;r)|_{r=1}\,,\\
\mathcal{P}_{t,\mathrm{mix}}&=\lim_{\delta\to0}\mathcal{H}_1(-\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;1).
\end{aligned}\tag{119}$$

The two separate soft-endpoint time integrals in Π_1 give equal completed integrals after their arguments and SK labels are reversed; this accounts for the factor 2 in the first line. All SK terms and both Whittaker indices are combined before removing the common regulator. The identity $\mathcal{H}_1(\alpha,\beta;1)^*=\mathcal{H}_1(\beta^*,\alpha^*;1)$ makes $\mathcal{P}_{t,\mathrm{mix}}$ real.

For a finite expression of the oscillatory coefficients, it is useful to retain one symmetric family of special functions,

$$\mathcal{Q}_{mn}(\tilde\nu)=\frac{2^{-1-2\mathrm{i}\tilde\nu-m-n}\Gamma(1+2\mathrm{i}\tilde\nu+m+n)}{(1+2\mathrm{i}\tilde\nu+m)(1+2\mathrm{i}\tilde\nu+n)}{}_3F_2\left[\begin{matrix}1+2\mathrm{i}\tilde\nu+m+n,\ 1+\mathrm{i}\tilde\nu,\ 1\\ 2+2\mathrm{i}\tilde\nu+m,\ 2+2\mathrm{i}\tilde\nu+n\end{matrix}\middle|1\right].\tag{120}$$

Only $\mathcal{Q}_{00},\mathcal{Q}_{01},\mathcal{Q}_{11},\mathcal{Q}_{02},\mathcal{Q}_{12}$ occur. The series has convergence balance $1+\mathrm{i}\tilde\nu$ and is convergent for real $\tilde\nu$. Reducing the full SK sums in (119) gives

$$\mathcal{P}_{t,1}=2\cosh(\pi\tilde\nu)\left[\mathcal{Q}_{00}+2\mathcal{Q}_{01}+\mathcal{Q}_{11}+\frac{2(\mathcal{Q}_{02}+\mathcal{Q}_{12})}{1+2\mathrm{i}\tilde\nu}\right]-\pi^2 2^{-1-2\mathrm{i}\tilde\nu}\frac{\Gamma(1+2\mathrm{i}\tilde\nu)^2}{\Gamma(1+\mathrm{i}\tilde\nu)^2},\tag{121}$$

$$\begin{aligned}
\mathcal{P}_{t,2}&=2\cosh(\pi\tilde\nu)\sum_{m,n=0}^{1}\left[\frac{m(1+2\mathrm{i}\tilde\nu+m)}{2(1+\mathrm{i}\tilde\nu+m)}+\frac{n(1+2\mathrm{i}\tilde\nu+n)}{2(1+\mathrm{i}\tilde\nu+n)}-(1+2\mathrm{i}\tilde\nu+m+n)\right]\mathcal{Q}_{mn}\\
&\quad+(1+2\mathrm{i}\tilde\nu)\pi^2 2^{-1-2\mathrm{i}\tilde\nu}\frac{\Gamma(1+2\mathrm{i}\tilde\nu)^2}{\Gamma(1+\mathrm{i}\tilde\nu)^2}\\
&\quad+2^{-1-2\mathrm{i}\tilde\nu}\cosh(\pi\tilde\nu)\Gamma(1+2\mathrm{i}\tilde\nu)\frac{4+5\mathrm{i}\tilde\nu-2\tilde\nu^2}{1+\mathrm{i}\tilde\nu}.
\end{aligned}\tag{122}$$

These expressions include the finite endpoint contributions of the assembled seeds. For the mixed coefficient the regulated expression in (119) provides the corresponding evaluation.

B.4 Signal Coefficients

With $\rho=1/2-\mathrm{i}\tilde\nu$, define the common soft factor

$$\mathcal{B}_t(\tilde\nu)=\frac{2^{1+4\mathrm{i}\tilde\nu}\Gamma(\frac{1}{2}-\mathrm{i}\tilde\nu)^4 f_0(\rho,\rho)}{32\pi^2(4\pi)^{3/2}(\rho-3)(\rho-2)(4\rho-7)(4\rho-5)}.\tag{123}$$

The factor includes the normalization of the two soft lines, the spin trace and the four orderings, together with the phase convention in (56): $8c_+^2\,16^{2\mathrm{i}\tilde{\nu}} = 2^{1+4\mathrm{i}\tilde{\nu}}\Gamma(1/2-\mathrm{i}\tilde{\nu})^4/\pi^2$. Substituting (110) gives

$$\begin{aligned}
A_{1,t} &= \mathcal{B}_t(\rho-1)(2\rho-3)\left[4(\rho-3)(3\mathcal{P}_{t,1}^2+2\mathcal{P}_{t,1}\mathcal{P}_{t,2})+(2\rho-5)\mathcal{P}_{t,2}^2\right],\\
A_{2,t} &= -\mathcal{B}_t\mathcal{P}_{t,2}^2(\rho-1)(2\rho-3)^2, \qquad A_{3,t} = -3A_{2,t},\\
B_t &= -2\mathcal{B}_t\mathcal{P}_{t,2}^2(\rho-1)(\rho-2)(2\rho-3),\\
D_t &= \mathcal{B}_t\mathcal{P}_{t,2}^2(\rho-2)(2\rho-5)(2\rho-3).
\end{aligned} \tag{124}$$

All couplings and the color factor remain outside these coefficients, as specified in (55).

For the opposite soft branches, the leading endpoint matrices are $L_{\mathrm{L}}^- = \mathbf{1}_2$, $L_{\bar{\mathrm{L}}}^- = -\boldsymbol{\epsilon}$, $U_{\mathrm{L}}^- = \boldsymbol{\epsilon}$ and $U_{\bar{\mathrm{L}}}^- = -\mathbf{1}_2$, with scalar prefactor $+c_-/2$. The vertex-type sums are

$$V_L^{-+,(0)} = F_{1;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\mathbf{1}_2, \qquad V_R^{+-,(0)} = F_{1;\mathsf{a}_4\mathsf{a}_3}(k_R;\tau_4,\tau_3)\mathbf{1}_2. \tag{125}$$

The superscripts here refer to the soft mass branches. The soft prefactor is $-|c_+|^2(q^2)^{-\mathrm{i}\tilde{\nu}}(K^2)^{\mathrm{i}\tilde{\nu}}$; its minus sign cancels the loop sign. There are two branch assignments and two external orderings at each end. Together with the Pauli trace these supply a factor 16. Using $|c_+|^2 = 1/[4\cosh^2(\pi\tilde{\nu})]$ and (59) gives the coefficient $2\tilde{\nu}(1+\tilde{\nu}^2)/[3\pi\sinh(2\pi\tilde{\nu})]$ in (60).

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